UME

Organisation Operating System

Design & Architecture Document

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Version	Date	Author	Changes
1.0.0	Jan 2025	Platform Engineering	Initial architecture document — OrgOS v1 release
2.0.0	Sep 2025	Platform Engineering	Major revision: ODD governance framework, SDO engine, AI OrgExecs
2.1.0	Nov 2025	Platform Engineering	Added DAO/DLT architecture, franchise network topology
2.2.0	Jan 2026	Platform Engineering	Multi-region deployment, OrgDNA versioning, security chapter
3.0.0	Mar 2026	Platform Engineering	Full platform v3 rewrite: OrgKernel v3, new module architecture, AI-native OrgExec registry

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Executive Summary

UME is a category-defining platform that delivers a fully integrated Organisation Operating System (OrgOS) — a single, unified platform on which any organisation can run every core business process, from accounting and payroll to AI agent governance, regulatory compliance, multi-entity legal structures, and decentralised autonomous organisation (DAO) operations.

This document provides the complete Design and Architecture specification for the UME platform, covering system architecture, module design, data models, integration patterns, security framework, deployment topology, and the novel governance primitives that distinguish UME from conventional enterprise software.

Platform Purpose

The central thesis of UME is that the fragmentation of business software — where organisations operate 15–40 disconnected SaaS tools — is the primary cause of operational inefficiency, governance failure, and data silos. UME replaces this fragmentation with a single, kernel-driven platform where every module, agent, device, and person operates under a unified operating contract framework called the Organisational Design Document (ODD).

Key Architectural Innovations

- OrgKernel — the central operating kernel that provides event routing, permission enforcement, and state management across all modules.
- Organisational Design Document (ODD) — a versioned, machine-readable governance contract that defines the capabilities, constraints, and operating rules for every actor in the organisation (human employees, AI agents, IoT devices).
- OrgDNA — the exportable, portable configuration of an entire organisation — its structure, module settings, ODD templates, and roles — enabling organisations to be cloned, franchised, or branched in under 60 seconds.
- Software-Defined Organisation (SDO) — a real-time digital twin of the organisation that models every entity, actor, and process as addressable objects with live state.
- OrgExec Registry — unified governance for all organisational actors regardless of type: human employees, AI language model agents, and IoT/OrgCPU devices all operate under the same ODD governance framework.
- DAO & DLT Layer — native support for decentralised autonomous organisation structures, on-chain governance voting, and smart contract royalty/revenue distribution across franchise and partner networks.

Document Scope

This document covers the complete UME platform as of version 3.0.0 (March 2026). It is structured into ten parts:

1. Platform Vision and Product Architecture
2. OrgKernel — Core Operating System
3. ODD Governance Framework
4. OrgDNA and Software-Defined Organisation
5. Module Architecture — all 12 platform modules
6. AI and OrgExec Architecture
7. DAO, DLT, and Smart Contract Architecture
8. Data Architecture and Storage
9. Integration and API Architecture
10. Security, Compliance, and Deployment Architecture

PART I

Platform Vision & Product Architecture

1. Platform Vision

1.1 The Problem: Organisational Software Fragmentation

The modern organisation operates in a state of permanent software fragmentation. A typical SMB uses between 12 and 22 SaaS applications for core business operations, while a mid-market firm uses 30–60. Each tool maintains its own data model, authentication system, permission framework, and integration surface. The result is:

- Data inconsistency: customer records, employee data, and financial figures exist in conflicting states across multiple systems.
- Governance gaps: no single policy engine governs all tools; each has its own admin console and compliance configuration.
- Integration tax: 15–30% of engineering time in digital-native businesses is spent maintaining integrations between tools that should be the same platform.
- AI agent incompatibility: modern AI agents and automation tools cannot operate safely across a fragmented tool stack without a governing permission layer.
- Franchise and multi-entity impossibility: scaling a business model across multiple legal entities or franchise networks requires bespoke software or expensive middleware that none of the fragment tools can provide.

1.2 The UME Answer: Organisation as Operating System

UME reconceptualises the organisation as a software system — an operating system — and builds a platform that implements that model comprehensively. Every business process is a module. Every person, agent, and device is a managed actor with an operating contract. Every event is logged, routed, and auditable. Every configuration is exportable and versionable.

> The UME mission: make running any organisation as reliable, configurable, and scalable as running a software system.

This is achieved through six foundational architectural choices:

Principle	Conventional SaaS	UME Platform
Data model	Fragmented across tools	Single unified OrgGraph data model
Governance	Per-tool admin consoles	ODD — single contract per actor, all tools
Actors	Only human users	Humans, AI agents, IoT devices — unified OrgExec model
Configuration	Manual per-tool settings	OrgDNA — versioned, exportable, cloneable
Multi-entity	Requires separate instances	Native multi-entity with consolidated views
Scalability	Add more tools	Add OrgApps to existing OrgKernel

1.3 Target Market Segments

UME is designed to serve five distinct market segments, each with a tailored entry point but sharing the same underlying platform:

Segment	Description	Primary Entry Point	Distinguishing Need
Solo & Micro	1–5 person businesses, freelancers	OrgOS template + Finance + CRM	Fast setup, invoicing, legal entity
SMB (Core)	5–50 employees, single entity	Full module stack, single site	Operational efficiency, compliance
Growth	50–250 employees, multi-site	Multi-entity, AI OrgExecs, GRC	Scale without headcount growth
Corporate / Enterprise	250+ employees, group structure	Group consolidation, ODD governance	Governance, audit, OrgDNA control
Franchise / DAO Network	Multi-org distributed network	OrgDNA clone, DAO layer, DLT	Consistent operations across entities

2. Platform Product Architecture

2.1 High-Level Architecture Overview

The UME platform is structured as a layered architecture with four principal tiers: the Infrastructure tier, the OrgKernel tier, the Module tier, and the Actor tier. Each tier has clearly defined responsibilities and interfaces.

UME Platform — 4-Tier Architecture

Tier	Layer Name	Responsibility	Key Components
Tier 4 (Top)	Actor Layer	All principals operating within the OrgOS	Human OrgExecs, AI Agents, OrgCPUs (IoT), External APIs
Tier 3	Module Layer	Domain-specific business functionality	12 OrgApp modules: Finance, HR, Legal, CRM, Ops, Risk, OrgSystem, DAO, Marketing, Work, Portal, Board
Tier 2	OrgKernel Layer	Core platform services — event bus, ODD enforcement, SDO, auth	OrgKernel, ODD Bus, SDO Engine, OrgDNA Manager, Audit Log, Event Bus
Tier 1 (Base)	Infrastructure Layer	Compute, storage, networking, observability	Kubernetes, PostgreSQL, Redis, Kafka, S3-compatible object store, Prometheus/Grafana

2.2 Module Catalogue

The UME platform ships with 12 first-party OrgApp modules. Additional modules can be installed from the OrgApp Marketplace or developed by third-party ISVs using the UME SDK.

Module	Code	Category	Core Capabilities
Finance & Accounting	FIN	Core	GL, AP/AR, invoicing, bank reconciliation, payroll, multi-entity consolidation, period close, budgeting

Module	Code	Category	Core Capabilities
HR & People	HR	Core	Employee records, ODD contracts, leave, payroll, performance, offboarding, OrgExec registry
Legal & Compliance	LEG	Core	Entity management, contract generation, filing calendar, IP registry, document vault, audit packs
Sales & CRM	CRM	Core	Customer records, order management, pipeline, invoicing integration, AI Comms agent
Operations & ERP	OPS	Core	Inventory, purchase orders, supplier management, equipment/OrgCPU registry, job management
Risk & GRC	RISK	Governance	Risk register, KRI monitoring, GRC framework, incident management, board reporting
OrgSystem	SYS	Platform	OrgApp registry, SDO digital twin, ODD editor, OrgDNA manager, audit trail, diagnostics
DAO & DLT	DAO	Extended	Decentralised governance, smart contracts, DLT ledger, royalty engine, franchise management
Marketing	MKT	Extended	Campaign management, micro-persona engine, AI content generation, attribution
Work Management	WORK	Extended	Tasks, sprints, projects, time tracking, OrgExec assignment
Employee Portal	EP	Extended	Self-service leave, payslips, expense claims, schedule view
Board Management	BRD	Governance	Board meetings, minutes, resolutions, director register, statutory reporting

2.3 Platform Editions

Edition	Modules Included	OrgExecs	Entities	AI Agents	DAO / DLT
Micro	Finance, CRM, HR (basic)	Up to 3	1	None	No
SMB	All Core modules	Up to 25	1	2 included	No

Edition	Modules Included	OrgExecs	Entities	AI Agents	DAO / DLT
Growth	All Core + Marketing + Work	Up to 100	Up to 3	5 included	No
Enterprise	All modules	Unlimited	Unlimited	Unlimited	Optional add-on
DAO Network	All modules + DAO & DLT	Unlimited	Unlimited	Unlimited	Yes — native

PART II

OrgKernel — Core Operating System

3. OrgKernel Architecture

3.1 Overview

The OrgKernel is the central processing and routing engine of the UME platform. Inspired by the design of operating system kernels, it provides process scheduling, resource management, inter-process communication, and security policy enforcement — but for organisational processes rather than computer processes.

Every interaction with the UME platform — whether initiated by a human user clicking a button, an AI agent executing a task, an OrgCPU sensor reporting a reading, or an external system calling an API — passes through the OrgKernel before reaching any module.

> The OrgKernel enforces a fundamental guarantee: no action can occur within the OrgOS without being authorised by the ODD of the acting OrgExec, logged on the audit trail, and emitted as an event on the event bus.

3.2 OrgKernel Components

Component	Role	Technology
ODD Enforcement Engine	Validates every action against the acting OrgExec's ODD before permitting execution	Rust policy engine (embedded)
Event Bus	Distributes all system events to subscribed modules and external integrations	Apache Kafka
State Manager	Maintains in-flight transaction state and ensures atomic multi-module operations	Redis (with PostgreSQL backing)
Audit Logger	Writes an immutable, append-only log of every event with full context	PostgreSQL append-only table + S3 archive
OrgExec Resolver	Resolves the identity and ODD of the acting principal on every request	Internal gRPC service

Component	Role	Technology
Module Router	Routes API requests and events to the correct OrgApp module	Internal service mesh (Istio)
SDO Engine	Maintains the Software-Defined Organisation digital twin in real time	Graph database (Neo4j)
OrgDNA Manager	Manages versioned OrgDNA snapshots and cloning operations	PostgreSQL + S3
Scheduler	Manages time-based triggers (filing deadlines, recurring jobs, cron events)	Temporal.io
Notification Bus	Routes notifications to human OrgExecs via email, SMS, in-app, and webhook	Twilio + SendGrid + internal WS

3.3 Request Lifecycle

Every request through the OrgKernel follows a deterministic six-stage lifecycle:

11. Authentication: The requesting principal is identified and their identity token is verified (JWT / mTLS for machine actors).
12. OrgExec Resolution: The ODD Enforcement Engine retrieves the full ODD contract for the principal and resolves their current permission state, including any time-bound or context-conditional grants.
13. Permission Check: The requested action is evaluated against the ODD. Denied actions are rejected with a structured error and logged. Granted actions proceed.
14. Action Execution: The action is routed to the target OrgApp module, which executes the business logic. The module returns a result object.
15. Event Emission: The OrgKernel emits one or more structured events to the Kafka event bus representing the state changes produced by the action.
16. Audit Write: A complete audit record is written atomically with the action commit, capturing: actor identity, ODD version, action type, target entity, parameters, timestamp, result, and correlation ID.

3.4 OrgKernel APIs

The OrgKernel exposes three API surfaces:

API Surface	Protocol	Consumers	Description
Public REST API	HTTPS REST / JSON	External integrations, third-party ISVs, mobile clients	Full CRUD API for all modules; versioned at /v1, /v2, /v3
Internal gRPC API	gRPC / Protobuf	Internal OrgApp modules, microservices	High-performance inter-service communication
GraphQL API	GraphQL over HTTPS	Dashboard frontends, reporting tools	Flexible read API for complex queries and SDO graph traversals
WebSocket API	WSS	Real-time UI, OrgCPU devices	Live event streaming, real-time dashboard updates, OrgCPU telemetry
Webhook Outbound	HTTPS POST	External systems, integration platforms	Push notifications for configured event types to external endpoints

3.5 Event Schema

All events on the UME event bus conform to a standard schema. This schema is the universal language of the UME platform and enables any module or external system to react to any event.

> UME Event Standard: every event contains a universally unique correlation ID, an ISO-8601 timestamp, the full identity of the acting OrgExec, the ODD version under which the action was authorised, the target entity, and the full before/after state delta.

Field	Type	Description
event_id	UUID v4	Globally unique event identifier
correlation_id	UUID v4	Links related events in a single transaction or workflow
org_id	String	The organisation (tenant) this event belongs to
actor_id	String	The OrgExec principal that triggered the event
actor_odd_version	Semver String	ODD contract version used to authorise this action
event_type	Enum String	e.g. finance.invoice.created, hr.leave.approved
module	Enum String	The OrgApp module that produced the event

Field	Type	Description
entity_type	String	e.g. Invoice, Employee, Order, RiskItem
entity_id	String	The target entity identifier
payload	JSON Object	Full before/after state delta of the entity
timestamp	ISO-8601	UTC timestamp of event occurrence
schema_version	Semver String	Event schema version — for forward/backward compatibility

PART III

ODD Governance Framework

4. Organisational Design Document (ODD)

4.1 What is an ODD?

An Organisational Design Document (ODD) is a machine-readable, versioned contract that formally defines the operating parameters of a single OrgExec within the organisation. An ODD is the central governance primitive of the UME platform — the concept that allows UME to govern all actors (human, AI, and device) under a unified framework.

An ODD answers four fundamental questions about any actor within the organisation:

- What can this actor see? (Data access and read permissions)
- What can this actor do? (Write operations and action permissions)
- Under what conditions? (Temporal, contextual, and threshold constraints)
- What happens when the actor cannot proceed? (Escalation and fallback rules)

4.2 ODD Architecture

An ODD is a JSON document stored in the UME data store with a full version history. Each ODD consists of five sections:

Section	Field Name	Description
Identity	identity	Actor ID, type (Human / AI Agent / OrgCPU), display name, department, entity scope
Capability Grants	capabilities	Explicit list of read/write/notify/execute permissions across all modules
Constraints	constraints	Operating limits: time windows, thresholds, rate limits, data access filters
Escalation Rules	escalation	Conditions under which the actor must request approval or escalate to a human supervisor

Section	Field Name	Description
Audit Configuration	audit	Audit verbosity level, data retention policy, monitoring alerts

4.3 ODD Capability Grant Reference

The following capability scopes are defined in UME v3. Each scope follows the format module:resource:action, conforming to the UME Permission Scope Standard (UPSS).

Module	Scope Pattern	Examples
Finance	fin:resource:action	fin:invoice:read, fin:invoice:write, fin:payroll:execute, fin:reports:read
HR	hr:resource:action	hr:employee:read, hr:employee:write, hr:leave:approve, hr:payroll:read
Legal	leg:resource:action	leg:contract:read, leg:filing:submit, leg:entity:read, leg:vault:write
CRM	crm:resource:action	crm:order:read, crm:order:write, crm:customer:read, crm:pipeline:write
Operations	ops:resource:action	ops:inventory:read, ops:po:write, ops:equipment:read, ops:schedule:write
Risk	risk:resource:action	risk:register:read, risk:kri:read, risk:incident:write, risk:board:read
OrgSystem	sys:resource:action	sys:odd:read, sys:odd:write, sys:sdp:read, sys:orgdna:export
DAO	dao:resource:action	dao:vote:cast, dao:treasury:read, dao:contract:deploy, dao:ledger:read

4.4 ODD Constraint Types

Constraint Type	Example	Enforcement Point
Temporal	working_hours: Mon-Fri 08:00-18:00	ODD Enforcement Engine — request time check

Constraint Type	Example	Enforcement Point
Threshold	max_invoice_value: 5000, approval_required_above: 10000	Action parameter validator
Rate Limit	max_actions_per_hour: 200 (AI agents)	Token bucket in Redis, enforced by ODD Engine
Entity Filter	data_scope: own_department_only	Query filter injected at module DB layer
Approval Gate	approve_required_for: payroll_run, period_close	Multi-signature workflow in OrgKernel
Escalation Trigger	escalate_if: order_queue_size > 50	SDO engine monitors KPI, triggers ODD escalation
Data Masking	mask_fields: [salary, national_insurance_no]	Response transformer in API gateway
Jurisdiction Scope	jurisdiction: england_wales	Legal module compliance rule filter

4.5 ODD Versioning and Lifecycle

Every ODD is semantically versioned following the format MAJOR.MINOR.PATCH. The version history of an ODD is immutable — once published, a version cannot be modified. New versions are created by proposing changes, reviewing the impact analysis (which the platform auto-generates), and approving the update.

Version Change	Trigger Examples	Approval Required
MAJOR	Removing a capability grant, reducing a threshold significantly, changing actor type	Yes — Owner or Director level OrgExec
MINOR	Adding a new capability, increasing a threshold, adding a new constraint	Yes — Manager level OrgExec
PATCH	Updating display metadata, adjusting notification preferences, changing audit level	Self-service — no approval needed

4.6 ODD and AI Agents

One of the most important applications of the ODD framework is AI agent governance. As organisations deploy large language model (LLM) agents and automation bots into their operations, the risk of ungoverned AI actions — executing financial transactions, modifying HR records, or communicating with customers without oversight — is a material operational and compliance risk.

UME's ODD framework addresses this risk by requiring every AI agent to have an ODD contract before it can be deployed within the OrgOS. An AI agent ODD has the same structure as a human ODD but typically includes:

- Strictly scoped capability grants (principle of least privilege).
- Rate limits to prevent runaway automation.
- Mandatory escalation triggers for high-impact actions.
- Audit verbosity set to maximum — every reasoning step logged.
- Hard kill switches: system-level capabilities to immediately revoke all grants and suspend the agent.

> The ODD framework provides the same compliance assurance for an AI agent as an employment contract provides for a human employee — defining exactly what the agent is authorised to do, and creating an immutable record of everything it does.

PART IV

OrgDNA & Software-Defined Organisation

5. OrgDNA — Portable Organisation Configuration

5.1 Concept and Purpose

OrgDNA is the exportable, portable representation of an entire organisation's configuration. It captures every structural and operational setting of an OrgOS instance in a single versioned artifact that can be stored, cloned, forked, compared, or broadcast to a network of affiliated organisations.

OrgDNA enables three transformative use cases:

- Organisation Cloning: A new franchise location, subsidiary, or spin-off can inherit the complete operational configuration of the parent organisation — including module settings, ODD templates, roles, chart of accounts, and compliance configurations — and be ready to operate in under 60 seconds.
- OrgDNA Broadcast: A franchisor or parent organisation can push configuration updates (such as new compliance rules, pricing structures, or ODD constraint changes) to all affiliated organisations simultaneously, with each affiliate able to accept or stage the update.
- Organisation Comparison: Two organisations sharing a common OrgDNA lineage can be compared to identify configuration drift — highlighting where an affiliate has deviated from the franchisor's standards.

5.2 OrgDNA Schema

An OrgDNA artifact is a structured JSON document compressed into a canonical UME archive format (.umegen). The schema has six root sections:

Section	Key	Contents
Identity	identity	Organisation name, legal entity type, jurisdiction, incorporation number, SIC codes, registration addresses
Structure	structure	Entity hierarchy, department tree, reporting lines, management structure

Section	Key	Contents
Module Configuration	modules	Enabled modules, feature flags, module-specific settings (e.g. chart of accounts, leave policy rules)
ODD Template Library	odd_templates	Standard ODD templates for all role types in the organisation — HR roles, AI agent roles, OrgCPU device roles
Role & Permission Map	roles	Named roles with associated ODD template references and module access configurations
Compliance Profile	compliance	Jurisdiction-specific compliance rules, filing obligations, VAT configuration, employment law settings

5.3 OrgDNA Versioning

OrgDNA follows the same semantic versioning scheme as ODD contracts. Each OrgDNA version is immutable after publication. The version history provides a complete audit trail of how the organisation's configuration has changed over time.

Change Type	MAJOR	MINOR	PATCH
Description	Breaking structural changes: entity restructuring, jurisdiction change, wholesale module reconfiguration	Additive changes: new roles, new ODD templates, new module features, compliance rule additions	Non-structural updates: metadata, display settings, notification preferences
Broadcast to Affiliates	Requires explicit affiliate acceptance with impact review	Auto-broadcast with 7-day acceptance window	Auto-applied to affiliates within 24 hours
Rollback	Supported with manual reconciliation wizard	Supported with automated rollback	Supported automatically

6. Software-Defined Organisation (SDO)

6.1 SDO Concept

The Software-Defined Organisation (SDO) is a real-time, queryable digital twin of the entire organisation. Every entity in the organisation — every employee, customer, supplier, order, invoice,

risk item, contract, and equipment asset — is represented as a node in the SDO graph. Relationships between entities are represented as typed edges.

The SDO graph is maintained in real time by the OrgKernel event bus. Every event that flows through the kernel updates the relevant nodes and edges in the SDO graph within milliseconds. This makes the SDO the single source of truth for the current state of the organisation.

> The SDO answers the question that no conventional ERP can answer in real time: "What is the complete, current, inter-connected state of the organisation at this exact moment?"

6.2 SDO Node Types

Node Type	Module	Key Properties	Key Relationships
Organisation	OrgSystem	Name, jurisdiction, incorporation, plan	Contains → Entities, Departments, OrgExecs
Entity	Legal	Legal name, company number, type, status	Owned by → Organisation; Has → Accounts, Contracts
OrgExec	HR/OrgSystem	Identity, type (Human/AI/Device), ODD version, status	Reports to → OrgExec; Has ODD → ODD Contract
Department	HR	Name, cost centre, manager	Contains → OrgExecs; Belongs to → Organisation
Invoice	Finance	Number, customer, amount, status, due date	Raised by → OrgExec; For → Customer; Posted to → GL Account
Order	CRM	Order ID, customer, items, stage	From → Customer; Assigned to → OrgExec; Generates → Invoice
Employee	HR	Name, role, pay rate, leave balance	Is an → OrgExec; In → Department; Has → ODD Contract
Risk	Risk	Description, impact, likelihood, status	Owned by → OrgExec; Linked to → KRI; In → Department
Contract	Legal	Type, parties, value, expiry	Between → Entities/Customers/Suppliers; Governed by → ODD
OrgCPU	Operations	Device ID, type, location, telemetry	Is an → OrgExec; Registered to → Entity; Has → ODD Contract

6.3 SDO Query API

The SDO graph is queryable via both GraphQL and a native Cypher-like UME Query Language (UQL). Common queries include:

- Traverse from a customer to all open orders, related invoices, and the OrgExecs assigned to those orders.
- Find all OrgExecs whose ODD contract expires within 60 days across all entities in the group.
- Identify all risks flagged HIGH that are owned by departments with below-90% compliance scores.
- Map the complete royalty flow from all franchise members to the parent DAO treasury.

PART V

Module Architecture

7. Module Architecture Overview

Each of the 12 UME OrgApp modules is an independently deployable service with its own data store, API surface, and event schema. However, modules are not isolated — they are tightly integrated through the OrgKernel event bus and share a unified data namespace through the SDO graph.

7.1 Module Design Principles

- Event-first: every module publishes all state changes as events before returning a response.
- ODD-aware: every module receives the acting OrgExec's resolved ODD from the OrgKernel and enforces it at the data layer.
- Stateless business logic: module business logic is stateless; all state is persisted to the module's own data store.
- Idempotent mutations: all write operations are idempotent and protected by distributed locks.
- Audit-by-default: all mutations emit an audit event before the transaction commits.

7.2 Module Internal Architecture

Each module follows a standard internal architecture with four layers:

Layer	Name	Responsibility
1 (API)	Module API Controller	Receives requests from OrgKernel router; validates input schema; calls service layer
2 (Service)	Business Service Layer	Implements domain business logic; enforces ODD constraints at application layer; calls repository
3 (Repository)	Data Repository	Abstracts database operations; implements ODD-driven data access filters; emits events

Layer	Name	Responsibility
4 (Store)	Module Data Store	PostgreSQL for relational data; Redis for caches and locks; S3 for documents and attachments

8. Finance & Accounting Module (FIN)

8.1 Module Purpose

The Finance & Accounting module is the financial core of the OrgOS. It provides a fully double-entry General Ledger (GL) combined with all operational financial tools — accounts payable, accounts receivable, invoicing, payroll, bank reconciliation, period close, and multi-entity consolidated reporting — in a single integrated system.

8.2 Core Sub-Systems

Sub-system	Description	Key Entities
General Ledger	Chart of accounts, journal entries, trial balance, period management	Account, JournalEntry, JournalLine, Period, TrialBalance
Accounts Receivable	Customer invoices, receipts, credit notes, AR ageing, collections	Invoice, Receipt, CreditNote, Customer, ARBalance
Accounts Payable	Supplier bills, payments, three-way match, AP ageing	Bill, Payment, Supplier, APBalance, PurchaseOrder
Payroll	Employee payroll, PAYE/NI calculation, pension, payslips, RTI submission	PayrollRun, PayrollLine, Payslip, TaxCode, PensionContribution
Bank Reconciliation	Statement import, transaction matching, reconciliation status	BankAccount, BankStatement, BankLine, Reconciliation
Period Management	Period open/close, pre-close checklist, lock enforcement	Period, PeriodClose, ClosingTask
Multi-Entity	Intercompany elimination, consolidated P&L/BS, group reporting	Group, ConsolidationRun, IntercompanyEntry
Budgeting	Budget planning, variance analysis, budget vs actuals	Budget, BudgetLine, Variance
Reporting Engine	P&L, Balance Sheet, Cash Flow, AR Ageing, AP Ageing, custom	Report, ReportTemplate, ReportRun

8.3 Chart of Accounts

The UME Finance module ships with jurisdiction-specific chart of accounts templates. The chart of accounts is organised in a five-tier hierarchy:

Tier	Code Range	Category	Examples
1	1000–1999	Assets	1100 Cash, 1200 Accounts Receivable, 1500 Fixed Assets
2	2000–2999	Liabilities	2100 Accounts Payable, 2200 PAYE Liability, 2400 Bank Loans
3	3000–3999	Equity	3100 Share Capital, 3200 Retained Earnings, 3300 Current Year Profit
4	4000–4999	Revenue	4100 Sales Revenue, 4200 Service Revenue, 4900 Other Income
5	5000–8999	Expenses	5100 Wages, 5200 Supplies, 6100 Rent, 7100 Insurance

8.4 Payroll Engine Architecture

The UME payroll engine is jurisdiction-aware and processes payroll in five computation stages:

17. Gross Pay Calculation: Hours worked × rate (hourly) or annual salary ÷ pay periods (salaried), plus any variable pay components.
18. Tax Code Resolution: Retrieves each employee's current tax code from the employee record and HMRC integration (UK), determining the applicable tax bands.
19. Deduction Computation: Calculates Income Tax, National Insurance (employee and employer contributions), pension contributions, student loan repayments, and any voluntary deductions.
20. Net Pay Derivation: Gross pay minus all deductions.
21. Payroll Journal Entry Generation: Automatically generates the complete set of double-entry journal entries for the payroll run — wages expense, tax liabilities, NI liabilities, pension liabilities, and net wages payable.

8.5 Finance Module Events

Event Type	Trigger	Key Subscribers
fin.invoice.created	New invoice saved/sent	CRM (order link), Audit, SDO, AR Balance updater

Event Type	Trigger	Key Subscribers
fin.invoice.paid	Payment received against invoice	CRM (order complete), AR Balance, Cash flow engine
fin.je.posted	Journal entry posted to GL	Trial Balance engine, Period manager, SDO, Audit
fin.payroll.processed	Payroll run approved and processed	HR (payslip generation), Legal (RTI submission), Audit
fin.period.closed	Accounting period locked	Compliance module, Reporting engine, SDO, Audit
fin.bank.reconciled	Bank account reconciled	Audit, SDO cash position updater

9. HR & People Module (HR)

9.1 Module Purpose

The HR & People module manages the complete lifecycle of all OrgExecs within the organisation — from initial onboarding and ODD contract creation through active employment management to structured offboarding. It is the registry of all human, AI agent, and OrgCPU actors and the primary interface for ODD contract management.

9.2 Core Sub-Systems

Sub-system	Description
OrgExec Registry	The definitive register of all actors (human, AI, device) with their current ODD contract, status, and identity
Employee Lifecycle	New hire wizard, probation management, role changes, promotion, termination, offboarding checklist
ODD Contract Manager	Create, version, review, and publish ODD contracts; impact analysis; approval workflow
Leave Management	Leave request, approval workflow, balance tracking, calendar integration, absence reporting
Payroll Integration	HR → Finance payroll data feed; payslip access portal; year-end P60/P11D generation
Performance Management	Review cycles, objectives setting, 360 feedback, rating history, PDPs
Organisation Chart	Live org chart derived from SDO; reporting line management; department management
Document Management	Employment contracts, offer letters, right-to-work documents, disciplinary records

9.3 OrgExec Type Matrix

The HR module manages three distinct OrgExec types. Each type has different ODD capabilities, audit requirements, and lifecycle processes:

Property	Human OrgExec	AI Agent OrgExec	OrgCPU (Device) OrgExec
Identity Source	HR record + SSO login	AI agent registry + API key	Device certificate + serial number
ODD Template Origin	Role-based HR templates	Agent function templates	Device class templates
Authentication	JWT (OAuth 2.0 / SAML SSO)	mTLS + API key rotation	mTLS + device certificate
Max Permissions	Full module access per role	Strictly scoped to agent function	Telemetry read + assigned ops scope
Audit Level	Standard	Maximum (all LLM reasoning logs)	Standard + sensor readings
Kill Switch	Account suspension	Immediate ODD revocation + API key invalidation	Device certificate revocation
Lifecycle Owner	HR Manager	OrgSystem Admin	Operations Manager

9.4 New Hire Wizard

The New Hire Wizard guides an HR manager through the complete onboarding process in four steps:

22. Step 1 — Personal Information: Name, contact details, role, department, start date, employment type, jurisdiction, right-to-work verification.
23. Step 2 — ODD Contract Configuration: Select OrgExec type, choose ODD template, customise capability grants and constraints, generate the ODD contract for approval.
24. Step 3 — Compensation Setup: Salary or hourly rate, tax code, pension scheme, bank details, benefit elections.
25. Step 4 — Review & Launch: Review all data, auto-generate employment contract (Legal module integration), send offer letter and access credentials, trigger system access provisioning via ODD activation.

9.5 Leave Management Rules Engine

Leave requests are processed through a configurable rules engine with the following default rule chain:

- Accrual Rate: Configurable per employment type and jurisdiction. Default UK full-time: 28 days statutory.
- Request Window: Minimum advance notice configurable (default: 2 weeks for holidays).
- Blackout Periods: Configurable periods during which leave cannot be taken (e.g. year-end for Finance team).
- Team Coverage Check: Validates that minimum team coverage (configurable threshold, default 60%) is maintained.
- Balance Check: Validates sufficient leave balance before approval.
- Auto-Approval: Leave requests meeting all rules can be configured for auto-approval below a threshold (default: 3 days or fewer).

10. Legal & Compliance Module (LEG)

10.1 Module Purpose

The Legal & Compliance module is the formal governance record-keeper of the organisation. It manages legal entities, regulatory filing obligations, contracts, intellectual property assets, and the document vault. Its compliance engine continuously monitors the organisation's obligations and surfaces upcoming deadlines.

10.2 Filing Obligation Engine

The filing obligation engine maintains a jurisdiction-specific library of regulatory obligations and maps them to the specific entities within the organisation. Key capabilities:

- Automatic filing calendar generation based on the organisation's jurisdiction, entity type, and financial year.
- Obligation-to-event linking: when a payment is made or a return is filed, the related obligation is automatically updated.
- Multi-jurisdiction support: organisations with entities in multiple jurisdictions receive correctly configured filing calendars for each entity.
- Advance notification: configurable notification triggers at 60, 30, 14, 7, and 1 days before each filing deadline.

10.3 Compliance Obligation Library

Jurisdiction	Obligation	Frequency	Module Integration
UK (England & Wales)	VAT Return (MTD)	Quarterly	Finance (Box 1–9 auto-calculation)
UK	PAYE RTI Monthly	Monthly	Finance Payroll (FPS/EPS submission)
UK	P11D Benefit in Kind	Annual	HR (benefit tracking)
UK	Annual Confirmation Statement	Annual	Legal entity registry
UK	Corporation Tax Return (CT600)	Annual	Finance (tax computation)
UK	Employer NIC	Monthly	Finance Payroll

Jurisdiction	Obligation	Frequency	Module Integration
US (Federal)	Federal Tax Deposit (EFTPS)	Semi-weekly / Monthly	Finance Payroll
US	Form 941 (Quarterly Payroll)	Quarterly	Finance Payroll
EU (VAT registered)	VAT Return	Monthly / Quarterly	Finance (OSS/IOSS support)
Multi-jurisdiction	Transfer Pricing Documentation	Annual	Finance Multi-entity

10.4 Contract Generation Engine

The contract generation engine creates legal documents using a template library that is pre-populated with data from the OrgOS. Supported contract types in the default library:

Contract Type	Auto-populated Fields	E-signature Integration
Employment Agreement (UK FTE)	Employer details, employee name/role/salary, start date, notice period, jurisdiction clauses	Yes — DocuSign / Adobe Sign
Employment Agreement (UK PT)	As FTE, plus hourly rate, contracted hours	Yes
Independent Contractor Agreement	Parties, scope of services, rate, IP assignment, termination	Yes
Non-Disclosure Agreement	Parties, scope of confidential information, term	Yes
Supplier Agreement	Supplier details, goods/services, payment terms, liability cap	Yes
Franchise Agreement	Franchisor/franchisee details, territory, royalty rate, standards reference	Yes — DAO smart contract linkage available
Partnership Agreement	Partners, profit share, decision-making rules, exit provisions	Yes

11. Sales & CRM Module (CRM)

11.1 Module Purpose

The Sales & CRM module manages the complete customer and order lifecycle — from initial contact and pipeline management through order fulfilment and collections. It is integrated with the Finance module for automatic invoice generation and with the Operations module for order fulfilment coordination.

11.2 Order State Machine

Orders in UME progress through a defined state machine. Each state transition emits an event and triggers the appropriate downstream actions:

State	Entry Trigger	Exit Trigger	Events Emitted	Downstream Actions
Draft	Order created	Customer confirmation received	crm.order.created	Notify assigned OrgExec
Received	Confirmation received	Work begun	crm.order.confirmed	Ops: add to queue; Comms AI: acknowledge customer
In Progress	OrgExec marks start	Work complete	crm.order.started	Ops: update equipment/resource allocation
Quality Check	Work marked complete	QC approved	crm.order.qc_pending	Notify QC OrgExec
Ready	QC approved	Customer collects or delivery dispatched	crm.order.ready	Comms AI: notify customer
Collected / Delivered	Handover confirmed	Invoice generated	crm.order.completed	Finance: auto-generate invoice
Invoiced	Invoice generated	Payment received	crm.order.invoiced	Finance: add to AR; Comms AI: send invoice
Closed	Payment received	Terminal state	crm.order.closed	Finance: mark paid; SDO: update customer LTV

State	Entry Trigger	Exit Trigger	Events Emitted	Downstream Actions
Cancelled	Cancellation request	Terminal state	crm.order.cancelled	Finance: reverse charges if applicable

11.3 Customer Micro-Persona Engine

The micro-persona engine builds a rich behavioural profile for each customer based on their transaction history, communication patterns, and preferences. This profile is used by the AI Comms Agent to personalise communications and by the Marketing module for campaign segmentation.

Persona Dimension	Data Sources	Use Cases
Order Frequency	CRM order history	Lapsed customer detection, re-engagement campaigns
Preferred Services	Order item analysis	Upsell recommendations, seasonal promotions
Communication Preference	Message open rates, response patterns	Channel selection for Comms AI (SMS vs email)
Price Sensitivity	Discount acceptance rate, order value trends	Promotion targeting, loyalty pricing
Lifecycle Stage	First order date, order frequency trend	New customer onboarding vs loyalty retention
Net Promoter Signal	Rating data, repeat referral patterns	Advocacy programme targeting

11.4 AI Comms Agent Integration

The Comms AI OrgExec is a specialised AI agent that manages outbound customer communications. It operates under a strict ODD contract that limits its permissions to outbound notifications only — it cannot modify any customer record or financial data.

- Automated order status notifications at each state transition.
- Personalised messaging using customer micro-persona data.
- Multi-channel delivery: SMS, email, in-app push notification.
- Escalation protocol: if a customer response requires a human decision, the Comms AI immediately escalates to the assigned OrgExec.

- All messages are logged on the audit trail with full attribution to the AI agent ODD.

12. Operations & ERP Module (OPS)

12.1 Module Purpose

The Operations module manages the physical and logistical backbone of the organisation — inventory, purchasing, supplier management, equipment (including IoT OrgCPU devices), and fulfilment operations. It provides the bridge between the commercial (CRM) and financial (Finance) layers and the physical production environment.

12.2 Inventory Management

Feature	Description
Multi-location inventory	Inventory tracked by location (site, bin, shelf) with real-time position updates
Reorder point management	Configurable minimum and reorder quantities per SKU; automatic low-stock alerts
Auto-PO generation	When stock falls below reorder point, a draft Purchase Order is auto-generated to the preferred supplier
Batch/lot tracking	Full batch and lot number tracking for regulated goods (food, chemicals, pharmaceuticals)
Inventory valuation	FIFO, LIFO, and weighted average cost methods; real-time valuation feeds to Finance GL
Consumption recording	OrgExecs (including AI agents) can record material consumption against orders

12.3 OrgCPU Device Integration

OrgCPU is the UME framework for integrating IoT devices, industrial equipment, and physical sensors into the OrgOS as first-class OrgExec actors. An OrgCPU device:

- Receives an ODD contract defining what data it can report and what operations it can trigger.
- Authenticates to the OrgKernel using a hardware-bound device certificate (mTLS).
- Reports telemetry data through the OrgKernel WebSocket API; data is stored in the Operations module.
- Can trigger operational events: e.g. a dry-cleaning machine completing a cycle emits ops.equipment.cycle_complete, which updates the associated order status in CRM.

- Receives commands through the OrgKernel command channel, subject to ODD approval: e.g. scheduling the next cycle, adjusting temperature settings.

12.4 Procurement Process Flow

The purchase-to-pay process in UME follows a standard workflow that integrates the Operations, Finance, and Legal modules:

26. Purchase Requisition: An OrgExec with ops:po:write capability raises a purchase requisition specifying supplier, items, quantities, and required delivery date.
27. Approval Routing: Requisitions above the ODD-configured threshold are automatically routed to the approving OrgExec. Below-threshold requisitions are auto-approved.
28. PO Generation: An approved requisition generates a formal Purchase Order, which is sent to the supplier electronically (PDF or EDI).
29. Goods Receipt: When goods are received, the receiving OrgExec records the goods receipt note (GRN) against the PO.
30. Three-Way Match: Finance module automatically matches the supplier invoice against the PO and GRN. Discrepancies are flagged for review. Matched invoices proceed to payment.
31. Payment Run: Matched invoices are included in the next payment run. Payment is processed and a payment journal entry is posted to the GL.

13. Risk & GRC Module (RISK)

13.1 Module Purpose

The Risk & GRC module provides the organisation with a structured framework for identifying, measuring, monitoring, and managing risk. It combines operational risk management, Key Risk Indicator (KRI) monitoring, governance controls assessment, and board-level reporting in a single integrated system connected to the live SDO data.

13.2 Risk Register Architecture

The risk register is the central inventory of all identified risks. Each risk is a structured record with the following properties:

Property	Description	Example Values
Risk ID	System-generated unique identifier	RISK-0042
Category	Risk taxonomy classification	Financial, Operational, Compliance, HR, IT, Strategic
Description	Plain-language description of the risk	"Key person dependency — only Alice can run payroll"
Impact	Potential business impact if risk materialises	Critical / High / Medium / Low
Likelihood	Probability of risk materialising	Certain / Likely / Possible / Unlikely / Rare
Inherent Score	Impact × Likelihood before controls	Critical (16), High (12), Medium (6), Low (2)
Controls	List of mitigating controls in place	Cross-training programme, documented procedures
Residual Score	Impact × Likelihood after controls	Reduced from inherent score
Risk Owner	OrgExec responsible for this risk	Alice L. (Manager)
Status	Current management status	Open / Mitigated / Accepted / Closed

Property	Description	Example Values
Review Date	Next scheduled review	2026-06-30

13.3 KRI Monitoring Engine

Key Risk Indicators (KRIs) are quantitative metrics that provide early warning signals of increasing risk levels. UME's KRI monitoring engine pulls live data from the SDO and evaluates configured thresholds in real time.

KRI	Data Source	Warning Threshold	Breach Threshold	Trigger Action
Cash Runway (days)	Finance — cash position / daily burn	45 days	30 days	Risk alert + board notification
AR Overdue > 60 days	Finance — AR ageing	£500	£1,000	Collections escalation + risk flag
Machine Uptime (%)	Operations — OrgCPU telemetry	85%	75%	Maintenance alert + ops escalation
Staff Absence Rate	HR — leave and absence	8%	12%	HR escalation + ops capacity review
Compliance Score	Legal — filing status + GRC	80%	70%	Legal alert + management notification
Order Cycle Time	CRM — order timestamps	+20% vs baseline	+40% vs baseline	Operations review + customer comms
Invoice Collection Rate	Finance — AR	< 85% within terms	< 70% within terms	Collections review + AR escalation

13.4 GRC Assessment Framework

The GRC (Governance, Risk, and Compliance) assessment framework provides a structured methodology for evaluating the organisation's control environment. UME ships with assessment templates aligned to common frameworks:

- ISO 27001 — Information Security Management System controls assessment
- COSO Internal Controls Framework — for financial reporting governance
- CIS Controls v8 — cybersecurity baseline assessment

- GDPR/UK GDPR — data protection compliance assessment
- Custom frameworks — organisations can define their own control libraries

14. OrgSystem Module (SYS)

14.1 Module Purpose

The OrgSystem module is the meta-management layer of the UME platform. It provides the interfaces for managing the platform itself: installing and configuring OrgApps, managing the OrgDNA, editing ODD contracts, monitoring the SDO digital twin, and accessing the system-wide audit trail. It is the "control plane" of the OrgOS.

14.2 OrgApp Management

OrgApps can be installed, configured, upgraded, or deactivated through the OrgSystem module. The OrgApp lifecycle:

32. Discovery: Browse the OrgApp marketplace or search for first-party modules.
33. Pre-install Assessment: The platform runs a pre-install assessment showing: compatibility with current OrgDNA version, ODD changes required, data migration requirements, and estimated setup time.
34. Installation: The OrgApp is deployed into the OrgKernel's module registry. Default ODD templates are added to the OrgDNA.
35. Configuration: Module settings are configured (chart of accounts for Finance, leave policies for HR, etc.).
36. OrgDNA Update: The OrgDNA is updated to include the new module's configuration, incrementing the MINOR version.
37. Activation: The module becomes active for all OrgExecs with the appropriate ODD grants.

14.3 Audit Trail Architecture

The UME audit trail is the immutable, append-only record of every event that has occurred within the OrgOS. It is designed to meet the strictest regulatory and forensic requirements:

Property	Specification
Storage	Append-only PostgreSQL table with WAL archiving to S3 (immutable object store)
Retention	Minimum 7 years (HMRC requirement); configurable up to indefinite for enterprise
Integrity	Each audit record is hash-chained to its predecessor; tampering is cryptographically detectable

Property	Specification
Granularity	Every read and write operation by every OrgExec is logged (configurable per ODD)
Export	SIEM-compatible export (CEF, JSON Lines); GDPR-compliant redaction tooling
Search	Full-text search with actor, time range, entity, event type, and correlation ID filters
Access Control	Audit trail is readable only by OrgExecs with sys:audit:read — typically Owner and Compliance Manager

15b. Marketing Module (MKT)

15b.1 Module Purpose

The Marketing module provides campaign management, customer segmentation, AI-powered content generation, and multi-channel campaign execution. It consumes customer micro-persona data from the CRM module to enable hyper-targeted communications.

15b.2 Campaign Architecture

Component	Description
Campaign Manager	Create, schedule, and track multi-channel marketing campaigns with audience segments, content, channel config, and attribution rules
Audience Segmentation Engine	Builds dynamic customer segments from CRM micro-persona data, order history, and engagement metrics
AI Content Generator	Uses LLM to generate personalised campaign copy, subject lines, and call-to-action variants (A/B testing)
Multi-Channel Executor	Dispatches campaigns across email (SendGrid), SMS (Twilio), in-app notifications, and print (PDF generation)
Attribution Engine	Tracks campaign-to-revenue attribution using UTM parameters, unique promo codes, and order source tagging in CRM
Analytics Dashboard	Open rates, click rates, conversion rates, revenue attribution, cost per acquisition

15c. Work Management Module (WORK)

15c.1 Module Purpose

The Work Management module provides task management, sprint planning, and project tracking capabilities. It is primarily used for internal team work — process improvement projects, compliance tasks, and operational initiatives — and integrates with all other modules to surface action items automatically.

Feature	Description
Task Management	Create, assign, prioritise, and track tasks; sub-tasks; recurring tasks; due dates; OrgExec assignment
Sprint / Kanban Board	Sprint planning with velocity tracking; Kanban board for continuous-flow teams
Project Templates	Pre-built project templates for common OrgOS activities: VAT return preparation, payroll run, machine maintenance, new hire onboarding
OrgOS Integration	Compliance filing deadlines auto-create tasks; KRI breaches auto-create risk mitigation tasks; new hire wizard auto-creates onboarding checklist
Time Tracking	Time logged against tasks; feeds to Finance for billable project cost allocation
Reporting	Sprint velocity, task completion rate, cycle time, blocked item analytics

15d. Board Management Module (BRD)

15d.1 Module Purpose

The Board Management module provides a governed workspace for company directors and board members. It manages board meeting packs, minutes, formal resolutions, the statutory director register, and compliance reporting required for board-level governance.

Feature	Description
Board Meeting Manager	Agenda builder, meeting pack distribution, minute recording, action tracking, attendance register
Formal Resolution Engine	Propose, circulate, vote on, and record formal resolutions; e-signature for directors; automatic filing to Legal vault
Director Register	Statutory register of directors, secretaries, and PSCs; Companies House synchronisation
Board Reporting Dashboard	Consolidated KPI pack from Finance, Risk, HR, and Compliance for board review
Conflict of Interest Register	Directors declare interests; conflicts checked automatically against proposed resolutions

Feature	Description
DAO Governance Integration	For DAO member organisations, DAO governance votes surfaced in Board module for formal ratification

PART VI

DAO, DLT & Smart Contract Architecture

15. DAO & DLT Module (DAO)

15.1 Concept: Organisation-as-DAO

The DAO & DLT module extends the UME OrgOS into the domain of decentralised autonomous organisations. It enables a network of legally independent organisations — such as a franchise network, a supply chain consortium, or an investment fund structure — to operate under shared governance rules enforced by on-chain smart contracts, with transparent financial flows and tamper-proof voting records.

The UME DAO architecture treats each participating organisation as a DAO member. Each member operates their own full UME OrgOS instance. The DAO layer provides the shared infrastructure that connects these instances:

DAO Layer Component	Purpose	Technology
DAO Registry	Maintains the list of member organisations, their roles, and participation status	PostgreSQL + DLT anchor
DLT Ledger	Distributed ledger providing tamper-proof record of all inter-organisation transactions and governance decisions	Hyperledger Fabric (permissioned) or Ethereum L2
Smart Contract Engine	Deploys and executes on-chain smart contracts for royalties, governance votes, and treasury distributions	Solidity (EVM) or Hyperledger Chaincode
Governance Engine	Manages proposals, voting rounds, and vote counting; enforces voting rules from the DAO charter	UME DAO Governance Service (Go)

DAO Layer Component	Purpose	Technology
Treasury Manager	Manages the shared DAO treasury; processes contributions and distributions	Smart contract + Finance module integration
OrgDNA Broadcast Service	Enables the DAO operator to broadcast OrgDNA updates to all member organisations	UME OrgDNA Manager + message bus

15.2 Smart Contract Architecture

UME ships with four standard smart contract templates that cover the most common DAO use cases:

Contract Template	Purpose	Key Parameters
Royalty Collection Contract	Automatically collects a percentage of monthly revenue from each member and distributes to the franchisor treasury	royalty_rate (%), collection_day, distribution_wallet, grace_period_days
Brand Standards Contract	Enforces that all member organisations remain within a specified OrgDNA version range	min_orgdna_version, max_major_drift, enforcement_action (notify / suspend)
Governance Charter Contract	Defines the rules for all governance votes: quorum, majority threshold, vote duration	quorum_pct, majority_pct, vote_duration_days, veto_rights
Treasury Distribution Contract	Distributes surplus from the treasury to member organisations according to configured shares	distribution_shares[], distribution_trigger (manual / scheduled / threshold), treasury_reserve_min

15.3 DLT Architecture: Permissioned vs Public

UME supports two DLT deployment modes, selected based on the organisation's requirements:

Property	Permissioned (Hyperledger Fabric)	Public EVM (Ethereum L2)
Access	Only DAO members can participate	Any address can interact
Privacy	Transaction details visible only to parties	Transactions public (amounts can be obscured)
Throughput	1,000+ TPS	50–500 TPS depending on L2
Finality	Immediate (Byzantine fault tolerant)	~12 seconds (L2 block time)
Cost	No gas fees; infrastructure cost only	Gas fees per transaction
Regulatory	Preferred for financial/regulated use cases	Preferred for open/public governance
Smart Contracts	Hyperledger Chaincode (Go/Java)	Solidity (ERC-20/ERC-721/custom)
UME Default	Yes — default for new DAO networks	Available as alternative configuration

15.4 DAO Governance Flow

The standard governance vote lifecycle in a UME DAO network:

38. Proposal Submission: Any member OrgExec with dao:proposal:write capability submits a governance proposal. The proposal is recorded on the DLT and broadcast to all member organisations.
39. Discussion Period: A configurable discussion period (default: 7 days) allows member organisations to review the proposal, request clarifications, and prepare their voting position.
40. Voting Round: The governance contract opens voting. Each member organisation casts their vote (For / Against / Abstain) through their UME DAO module. Votes are committed to the DLT.
41. Quorum Check: At the close of the voting period, the governance contract checks whether the configured quorum (default: 60% of members must vote) was reached.
42. Result Determination: If quorum is met, the contract determines the result based on the configured majority threshold. Simple majority (>50%), supermajority (>67%), or unanimity.
43. Execution: Approved proposals that have corresponding smart contract implementations are automatically executed. OrgDNA updates are broadcast. Financial distributions are processed through the treasury contract.

PART VII

AI & OrgExec Architecture

16. AI Agent Architecture

16.1 AI Agent Design Principles

UME's AI agent framework is built on four design principles that distinguish it from conventional AI automation tools:

- Governed by contract, not by prompt: Every AI agent operates under an ODD contract, not a system prompt. The ODD is the authoritative source of what the agent can and cannot do — the LLM prompt only describes how to do it.
- Least privilege by default: AI agent ODDs are configured with the minimum permissions required for the agent's function. Agents cannot self-escalate permissions.
- Auditable to the action level: Every action an AI agent takes — including the reasoning trace that led to it — is logged on the immutable audit trail.
- Human-in-the-loop by design: Every AI agent has configurable escalation thresholds above which it must request human approval before proceeding.

16.2 AI Agent Runtime Architecture

AI agents in UME are hosted as containerised microservices within the OrgKernel service mesh. Each agent consists of:

Component	Description	Technology
Agent Runtime Container	Containerised process hosting the LLM inference engine and tool-calling framework	Docker on Kubernetes; GPU-accelerated nodes for on-premises inference
LLM Integration Layer	Supports pluggable LLM backends: OpenAI GPT-4/o, Anthropic Claude, Google Gemini, or on-premises models	LiteLLM abstraction layer; model router
Tool Registry	Declares the specific UME API tools the agent is authorised to call,	Tool registry service; ODD-aware tool filter

Component	Description	Technology
	cross-referenced with ODD capability grants	
Memory Manager	Manages short-term (context window) and long-term (vector store) memory for the agent	In-context + Pinecone/pgvector for long-term
Action Validator	Pre-validates every tool call against the ODD before it is sent to the OrgKernel	ODD Enforcement Engine embedded client
Escalation Handler	Detects escalation conditions and routes to the configured human OrgExec via the notification bus	UME Escalation Service
Reasoning Logger	Captures the full LLM reasoning trace (chain of thought) for every action and logs it to the audit trail	Structured logging to Audit Logger

16.3 First-Party AI Agents

UME ships with three first-party AI agents included in the Growth plan and above:

Agent	Function	Key Tools	Default ODD Constraints
Scheduling Agent	Optimises daily order scheduling, staff allocation, and equipment utilisation using demand forecasting	crm:order:read, hr:schedule:write, ops:equipment:read, ops:schedule:write	Max 200 schedule changes/hr; escalate if demand +40% vs forecast; 24/7 operating hours
Customer Comms Agent	Sends personalised order status notifications, responds to standard customer enquiries, manages review requests	crm:order:read, crm:customer:read, notify:sms, notify:email	Outbound comms only — cannot modify any record; escalate on any complaint or claim; max 500 messages/day
Compliance Monitor Agent	Continuously monitors filing calendar, KRI thresholds, and ODD	risk:kri:read, leg:filing:read, hr:odd:read,	Read-only access to all modules; no write

Agent	Function	Key Tools	Default ODD Constraints
	expiry dates; proactively alerts responsible OrgExecs	sys:audit:read, notify:internal	permissions; escalation to Owner on Critical risk breach

16.4 Custom AI Agent Development

Organisations can develop and deploy custom AI agents using the UME Agent SDK. The SDK provides:

- ODD contract scaffolding: generate a starting ODD template for a custom agent based on the specified function.
- Tool registry integration: register custom tools that call any UME API endpoint, with automatic ODD capability mapping.
- Testing harness: simulate agent actions in a sandbox environment against a test OrgOS instance before production deployment.
- Monitoring integration: custom agents automatically appear in the OrgSystem AI OrgExec registry with standard health monitoring.

PART VIII

Data Architecture

17. Data Architecture

17.1 Multi-Tenancy Model

UME implements a row-level tenant isolation model within a shared PostgreSQL cluster. Every table in every module database includes an org_id column which is enforced at the Row Security Policy (RSP) level. This means:

- A single database cluster serves all tenants, providing economies of scale.
- Row Security Policies ensure it is architecturally impossible for one tenant's query to return another tenant's data.
- Enterprise customers with regulatory requirements (e.g. financial services, healthcare) can be provisioned on dedicated database clusters at no change to the application layer.
- DAO network members each have their own org_id scope; inter-organisation data sharing is mediated exclusively through the DAO layer APIs.

17.2 Data Store Responsibilities

Data Store	Technology	Purpose	Modules
Primary RDBMS	PostgreSQL 16	All transactional data — accounts, employees, orders, events, audit trail	All modules
Graph Database	Neo4j 5	SDO digital twin — nodes and relationships; real-time org graph queries	OrgSystem (SDO engine)
Cache Layer	Redis 7	Session state, ODD resolved permissions, API rate limiting, distributed locks	OrgKernel, all modules

Data Store	Technology	Purpose	Modules
Event Streaming	Apache Kafka 3.7	Asynchronous event bus between modules; event replay and audit event stream	OrgKernel (all events)
Document Store	S3-compatible (MinIO / AWS S3)	Contracts, payslips, attachments, audit archive, OrgDNA snapshots	Legal, HR, Finance, OrgSystem
Search Index	Elasticsearch 8	Full-text search across all modules; customer search, document search, audit search	All modules
Time-Series DB	TimescaleDB (PostgreSQL extension)	OrgCPU telemetry data, KRI historical time-series, performance metrics	Operations, Risk
Vector Store	pgvector (PostgreSQL extension)	AI agent long-term memory, semantic document search, micro-persona embeddings	AI agents, CRM, Marketing
Workflow Engine	Temporal.io	Long-running workflows, scheduled jobs, approval workflows, ODD lifecycle	OrgKernel, all modules

17.3 Data Retention Policy

Data Category	Retention Period	Legal Basis	Deletion Method
Financial records (GL, invoices, payroll)	7 years from transaction date	HMRC requirement (UK); IRS (US)	Hard delete after retention expires
Employee records (active)	Duration of employment + 7 years	Employment law + HMRC	Anonymise personal fields; retain for HMRC
Audit trail records	7 years minimum; configurable up to indefinite	Regulatory + forensic	Hash-chained; cannot be deleted

Data Category	Retention Period	Legal Basis	Deletion Method
Contract documents	Contract term + 7 years	Statute of limitations	Vault archive; access-restricted after expiry
ODD contract history	Indefinite — immutable governance record	OrgOS governance policy	Not deletable
Customer personal data	Order history retention period (default 3 years) + legal claims window	UK GDPR / CCPA	Right-to-erasure tooling available
AI agent reasoning logs	2 years (configurable)	OrgOS AI governance policy	Automated purge after retention
Application logs	90 days	Operational	Log rotation and deletion

17.4 Database Schema Conventions

All UME database tables follow a consistent schema convention:

- Primary Key: UUID v4 (id column) — globally unique, not sequential, safe for distributed generation.
- Tenant Isolation: org_id VARCHAR NOT NULL with Row Security Policy.
- Soft Delete: deleted_at TIMESTAMPTZ NULL — records are soft-deleted by setting this field; a hard-delete background job runs after the retention period.
- Optimistic Locking: version INT NOT NULL DEFAULT 1 — incremented on every update; used for conflict detection.
- Audit Timestamps: created_at, updated_at TIMESTAMPTZ NOT NULL — automatically maintained.
- Actor Attribution: created_by_actor_id, updated_by_actor_id UUID — references the OrgExec who created/modified the record.
- ODD Version Snapshot: acted_under_odd_version VARCHAR — the ODD contract version in effect when the record was created/modified.

PART IX

Integration Architecture

18. Integration Architecture

18.1 Integration Philosophy

UME is designed to be the system of record for all core business operations. Its integration strategy therefore focuses on two scenarios: (1) consuming data from specialised external systems that UME does not replace (e.g. payment gateways, government e-filing portals), and (2) providing data to downstream systems that need access to UME data (e.g. business intelligence tools, legacy systems).

UME explicitly does not build integrations designed to maintain data in parallel with another system of record — such integrations perpetuate the fragmentation problem UME is designed to solve.

18.2 First-Party Integration Catalogue

Category	Integration	Direction	Module	Description
Payments	Stripe	Inbound (webhooks)	Finance / CRM	Payment received events → auto-match invoices; Stripe payment links on invoices
Payments	GoCardless	Inbound	Finance	Direct debit collection; recurring billing; payment status webhooks
Banking	Open Banking (PSD2)	Inbound	Finance	Bank statement import via Open Banking APIs; automatic reconciliation feed
Tax (UK)	HMRC Making Tax Digital	Outbound	Finance / Legal	VAT return submission; PAYE RTI submission; Corporation Tax data preparation
E-signature	DocuSign	Bidirectional	Legal	Contract sent for signing; signed document webhook → Legal vault

Category	Integration	Direction	Module	Description
E-signature	Adobe Sign	Bidirectional	Legal	Alternative e-signature provider
Communication	Twilio SMS	Outbound	CRM / Comms AI	Customer SMS notifications; 2FA SMS codes
Communication	SendGrid	Outbound	All modules	Transactional email — invoices, notifications, payslips
Calendar	Google Calendar / Outlook	Bidirectional	HR / Work	Leave calendar sync; meeting booking for board management
Accounting	Xero Import	Inbound (one-time)	Finance	Historical data migration from Xero during onboarding
Accounting	QuickBooks Import	Inbound (one-time)	Finance	Historical data migration from QuickBooks
Identity	Google Workspace SSO	Inbound	OrgKernel	SAML 2.0 SSO; user provisioning via SCIM
Identity	Microsoft Entra ID	Inbound	OrgKernel	SAML 2.0 SSO; user provisioning via SCIM
IoT	MQTT Broker	Inbound	Operations	OrgCPU device telemetry ingestion via MQTT protocol

18.3 Webhook Architecture

UME provides a configurable outbound webhook system for integration with external platforms. Webhooks can be configured to fire on any event type from the UME event schema.

Property	Specification
Delivery	HTTPS POST to configured endpoint; retry on failure (exponential backoff, max 5 retries over 24 hours)

Property	Specification
Payload Format	JSON; conforms to UME event schema; configurable field selection
Security	HMAC-SHA256 signature on every request; rotating signing keys; configurable IP allowlist
Ordering	Best-effort ordering with correlation_id for reconstruction; not guaranteed strictly ordered
Filtering	Subscribe to specific event types, specific entities, or specific actor types
Testing	Test mode sends synthetic events to configured endpoint; webhook log available in OrgSystem

18.4 API Rate Limits

Plan	API Requests/minute	Burst Limit	Webhook Deliveries/day	Concurrent WebSocket Connections
Micro	60	120 for 30 seconds	1,000	5
SMB	300	600 for 60 seconds	10,000	25
Growth	1,000	3,000 for 60 seconds	100,000	100
Enterprise	10,000	50,000 for 60 seconds	Unlimited	Unlimited
DAO Network (per member)	2,000	6,000 for 60 seconds	200,000	200

PART X

Security Architecture

19. Security Architecture

19.1 Security Model

The UME security model is built on three layers: network-level security (infrastructure perimeter), transport-level security (all communications encrypted), and application-level security (ODD-enforced permission model). The application-level security layer is uniquely powerful because it enforces permissions not just at the API gateway but at the data layer — meaning even a compromised internal service cannot access out-of-scope data.

19.2 Authentication Architecture

Actor Type	Authentication Method	Token Type	Token Lifetime	MFA
Human OrgExec (browser)	Username/password or SAML SSO	JWT (short-lived) + Refresh token	15 min / 30 days	Required for Owner/Admin; optional for others (configurable)
Human OrgExec (mobile)	Biometric + PIN via PKCE flow	JWT + Refresh	15 min / 7 days	Biometric = MFA
AI Agent	API key + mutual TLS	API key (rotating)	24 hours (auto-rotated)	N/A — mTLS provides second factor
OrgCPU Device	Hardware device certificate + serial number	Device cert (mTLS)	1 year (renewed automatically)	N/A — hardware cert
External Integration	OAuth 2.0 Client Credentials + API key	OAuth access token	1 hour	N/A
DAO Smart Contract	Wallet signature (ECDSA)	Signed transaction	Per-transaction	Hardware wallet option

19.3 Encryption Specifications

Data Category	At Rest	In Transit	Key Management
All database data	AES-256-GCM (storage-level encryption)	TLS 1.3	AWS KMS / HashiCorp Vault
Document vault (contracts, payslips)	AES-256-GCM + field-level encryption for PII	TLS 1.3	Per-document encryption keys in KMS
Personal data fields	Column-level AES-256 with per-tenant keys	TLS 1.3	Tenant-specific key rotation
API communications	N/A	TLS 1.3 minimum; TLS 1.2 deprecated	Certificate managed by platform
Internal service mesh	N/A	mTLS (Istio)	Istio CA / cert-manager
Device telemetry (OrgCPU)	AES-256 on device	mTLS	Device-provisioned certificate
DLT transactions	ECDSA signatures on all transactions	TLS 1.3 for node communication	Organisation wallet (custodial or non-custodial)
Backups	AES-256-GCM	TLS for backup transfer	Separate backup encryption keys in KMS

19.4 Vulnerability Management

UME operates a continuous vulnerability management programme:

- Automated dependency scanning on every build (Dependabot + Snyk) with critical/high vulnerabilities blocking deployment.
- Weekly SAST (static application security testing) via CodeQL across all repositories.
- Quarterly penetration tests by an independent third-party security firm.
- Bug bounty programme (HackerOne) for responsible disclosure of security vulnerabilities.

- Zero-trust network architecture — no implicit trust between services; all inter-service calls authenticated via mTLS.
- Security incident response plan with defined SLAs: Critical (2-hour response), High (8-hour), Medium (48-hour), Low (7-day).

19.5 GDPR and Data Privacy Compliance

GDPR Requirement	UME Implementation
Lawful basis for processing	Configurable per-data-category legal basis; documented in Privacy Policy module
Right of access (SAR)	Automated Subject Access Request export tool: generates complete JSON of all data for a natural person across all modules
Right to erasure	GDPR Erasure Wizard: anonymises personal fields in all tables, removes documents, marks for audit trail — within 30 days
Data minimisation	ODD-level field-level read permissions enforce data minimisation; AI agents cannot access fields not in their ODD
Portability	Data export API returns all customer/employee data in machine-readable JSON format
Privacy by design	Column-level encryption on all PII fields; data access logged; no data shared between tenants
DPA / Sub-processors	DPA available; sub-processor list published; security reviews on all sub-processors
Data residency	EU-only deployment available; UK-only deployment available for UK GDPR compliance

PART XI

Deployment Architecture

20. Deployment Architecture

20.1 Deployment Models

UME supports three deployment models to accommodate different organisational requirements:

Model	Description	Target Customer	SLA
UME Cloud (SaaS)	Fully managed multi-tenant cloud deployment on UME infrastructure. Zero ops overhead for the customer.	Micro, SMB, Growth, most Enterprise	99.95% uptime; < 200ms p95 API response
UME Dedicated	Single-tenant deployment on UME-managed infrastructure. Dedicated compute, dedicated database cluster, customer-controlled encryption keys.	Enterprise with data residency requirements	99.99% uptime; < 150ms p95
UME On-Premises	Customer-hosted deployment on customer infrastructure (Kubernetes). UME provides Helm charts and platform engineering support.	Highly regulated industries (defence, finance, healthcare)	Customer-defined; UME SLA for software updates only

20.2 Cloud Architecture (SaaS)

The UME Cloud deployment runs on a multi-region Kubernetes infrastructure with active-active high availability across two primary regions and a standby disaster recovery region.

Component	Technology	Scaling Model	HA Configuration
API Gateway	Kong + Nginx	Horizontal auto-scale; minimum 3 replicas	Active-active across AZs
OrgKernel Services	Go microservices on Kubernetes	Horizontal auto-scale; HPA on CPU/RPS	Active-active; stateless

Component	Technology	Scaling Model	HA Configuration
OrgApp Module Services	Node.js / Go on Kubernetes	Per-module HPA; scale to zero for inactive modules	Active-active; stateless
PostgreSQL	Managed PostgreSQL (RDS / Cloud SQL)	Vertical + read replicas	Multi-AZ primary/replica; auto-failover < 30s
Redis	Redis Cluster (ElastiCache)	Horizontal sharding	Multi-AZ replication
Kafka	MSK (AWS) / Confluent	6-broker cluster; replication factor 3	Multi-AZ; 48-hour retention
Neo4j	Neo4j AuraDB	Managed scaling	Causal cluster; 3 nodes
S3	AWS S3 / GCS	Unlimited	Cross-region replication
CDN / Frontend	CloudFront / Cloudflare	Auto-scale	Multi-PoP global distribution

20.3 Regional Availability

Region	Cloud Provider	Data Residency	Available Plans
EU West (Ireland)	AWS eu-west-1	EU data residency (GDPR compliant)	All plans
EU Central (Frankfurt)	AWS eu-central-1	EU data residency; DSGVO compliant	All plans
UK (London)	AWS eu-west-2	UK data residency (UK GDPR)	All plans
US East (N. Virginia)	AWS us-east-1	US data residency	All plans
US West (Oregon)	AWS us-west-2	US data residency	All plans
AP Southeast (Singapore)	AWS ap-southeast-1	APAC data residency	Growth + Enterprise

Region	Cloud Provider	Data Residency	Available Plans
AU (Sydney)	AWS ap-southeast-2	Australian data residency	Enterprise + DAO Network

20.4 Disaster Recovery

Tier	RTO	RPO	Method
Transaction data (PostgreSQL)	< 30 seconds	< 1 second	Synchronous multi-AZ replication with automatic failover
Event data (Kafka)	< 60 seconds	< 5 seconds	Cross-AZ Kafka mirroring; consumer offset checkpointing
Graph data (Neo4j)	< 5 minutes	< 30 seconds	Neo4j causal cluster with cross-AZ secondary
Document store (S3)	< 5 minutes	Near-zero	Cross-region S3 replication with versioning
Full platform (region failure)	< 15 minutes	< 60 seconds	Cross-region active-passive; automated DNS failover
On-premises deployment	Customer-defined	Customer-defined	Backup scripts included; cold standby Helm deployment

PART XII

Performance & Observability

21. Performance Architecture

21.1 Performance Targets

Metric	Target	Measurement Method
API Response Time (p50)	< 80ms	Prometheus histogram; measured at API gateway
API Response Time (p95)	< 200ms	Prometheus histogram
API Response Time (p99)	< 500ms	Prometheus histogram
Dashboard Load Time (full)	< 1.2 seconds	Synthetic monitoring (Playwright); Real User Monitoring
Event Bus Latency (p95)	< 50ms producer → consumer	Kafka consumer lag monitoring
SDO Graph Query (p95)	< 100ms for 3-hop traversal	Neo4j query log monitoring
Database Query (p95)	< 20ms for indexed queries	pg_stat_statements monitoring
Audit Write Latency	< 5ms (non-blocking)	Prometheus histogram
ODD Enforcement (p99)	< 10ms	Internal timing middleware
AI Agent Tool Call Round-Trip	< 800ms (p95)	Agent telemetry

21.2 Caching Strategy

Cache Layer	What is Cached	TTL	Invalidation
ODD Resolved Permissions	Fully-resolved permission set for each authenticated OrgExec	5 minutes	Event-driven: invalidated on ODD update
SDO Node Cache	Frequently-accessed SDO nodes (top 10% by access rate)	30 seconds	Event-driven: invalidated on entity update event
Chart of Accounts	Full COA tree per org	10 minutes	Invalidated on account create/update
KRI Current Values	Latest KRI readings for dashboard display	60 seconds	Event-driven on source data change
Module Config	Module settings and feature flags per org	5 minutes	Invalidated on OrgDNA update
User Session	Authenticated session + decoded JWT claims	15 minutes (JWT TTL)	Logout or JWT expiry
API Response Cache	Read-only API responses for heavy list queries (invoice list, order history)	30 seconds	Event-driven on mutation events

21.3 Observability Stack

Concern	Tool	Key Metrics / Data
Metrics	Prometheus + Grafana	API latency, error rates, DB query times, Kafka consumer lag, Redis hit rates, ODD enforcement timing
Tracing	OpenTelemetry + Jaeger	Distributed trace for every request; span per service hop; ODD enforcement span; database query spans
Logging	Structured JSON logs → Loki → Grafana	Request logs, error logs, audit events, AI agent reasoning logs
Alerting	Alertmanager + PagerDuty	SLA breach alerts, error rate spikes, Kafka lag alerts, security events, KRI monitoring alerts

Concern	Tool	Key Metrics / Data
Synthetic Monitoring	Playwright (UME Canary)	End-to-end user journey tests run every 5 minutes from 6 global locations
RUM	OpenTelemetry Web SDK	Core Web Vitals, JS error rates, user journey completion rates from real browsers
Uptime	UptimeRobot + StatusPage	Public status page with per-module status and incident history

PART XIII

Developer Platform & SDK

22. Developer Platform

22.1 SDK Overview

The UME Developer Platform enables third-party developers and in-house engineering teams to extend the UME platform with custom OrgApps, custom AI agents, custom ODD templates, and custom integrations. The SDK is available for TypeScript/JavaScript (primary), Python, and Go.

22.2 SDK Capabilities

SDK Module	Language	Capabilities
@ume/sdk-core	TypeScript / JS	Authentication, OrgKernel API client, event subscription, ODD management, entity CRUD
@ume/sdk-agent	TypeScript / Python	AI agent scaffolding, tool registry, ODD-aware tool filter, escalation handler, memory manager
@ume/sdk-orgapp	TypeScript	OrgApp module scaffolding, module event schema, module API registration, OrgDNA integration
@ume/sdk-dlt	TypeScript / Go	Smart contract deployment, DLT event subscription, wallet integration, DAO governance client
@ume/sdk-orgcpu	C / Python / Go	OrgCPU device authentication, telemetry reporting, command handler, ODD device client
ume-go	Go	Full API client for Go — same capabilities as @ume/sdk-core
ume-python	Python	Full API client for Python — ideal for data science, analytics, and AI agent use cases

22.3 OrgApp Development Lifecycle

44. Scaffold: Use the UME CLI (ume init --orgapp) to generate the OrgApp project structure, including module manifest, event schema, ODD capability definitions, and API controller stubs.
45. Develop: Implement business logic in the service layer. Register event handlers for OrgKernel events. Define the ODD capability scopes your module will require from acting OrgExecs.
46. Test: Deploy the OrgApp to the UME Developer Sandbox — a full isolated OrgOS instance provided free of charge for development. Run integration tests using the SDK test harness.
47. Package: Build the OrgApp into a UME Package (.umepkg) — a signed, versioned bundle including the OrgApp manifest, Docker image, ODD templates, and migration scripts.
48. Publish: Submit to the UME OrgApp Marketplace for review (for public apps) or deploy directly to the organisation's OrgOS (for private apps).
49. Install: Organisations install the OrgApp through the OrgSystem module. The OrgApp is registered in the OrgKernel module registry and its ODD templates are added to the OrgDNA.

PART XIV

Platform Roadmap

23. Platform Roadmap

23.1 Current Version — v3.0.0 (March 2026)

Version 3.0.0 is the current production release of the UME platform. Key capabilities delivered in this version:

- OrgKernel v3 — complete rewrite with improved event throughput (5× v2), sub-10ms ODD enforcement, and native multi-region support.
- AI OrgExec Registry — unified governance for AI agents with full ODD framework, LLM model abstraction, and reasoning audit logging.
- DAO & DLT Module GA — production-ready DAO layer with both Hyperledger Fabric and EVM support.
- OrgDNA v3 — full OrgDNA schema with franchise broadcast, version diff tooling, and impact analysis.
- OrgCPU Framework — IoT device OrgExec registration, mTLS authentication, and MQTT telemetry ingestion.

23.2 Near-Term Roadmap — v3.1–3.3 (Q2–Q4 2026)

Version	Quarter	Theme	Key Features
v3.1	Q2 2026	AI-Native Operations	OrgExec Co-Pilot (AI assistant for all module screens); Predictive cash flow forecasting; AI-assisted contract drafting in Legal module
v3.2	Q3 2026	Global Expansion	US payroll (multi-state); EU VAT OSS support; Australia BAS filing; Singapore GST. New regional data centres: UAE, Brazil
v3.3	Q4 2026	Enterprise Controls	SOC 2 Type II certification; FedRAMP Ready (US gov); Advanced GRC — SOX-ready controls; Group audit trail across all entities

23.3 Medium-Term Roadmap — v4.0 (2027)

Theme	Description
OrgOS Marketplace	Full third-party OrgApp marketplace with ISV SDK, Stripe-powered revenue share, and OrgApp review/certification process
Agentic OrgOS	Fully agentic mode: OrgOS can be configured to have an AI Chief of Staff agent that orchestrates other AI agents and surfaces decisions to human OrgExecs proactively
Multi-Org Consortium	Supply chain consortium DAO — multiple independent organisations (including competitors) operating shared infrastructure with privacy-preserving data sharing via ZK proofs
OrgDNA NFT	Organisation configuration as an NFT on a public blockchain — enables OrgDNA transfer, acquisition, and inheritance as a verifiable on-chain asset
Voice OrgExec Interface	Voice-activated OrgOS operations — query the SDO, raise invoices, approve leave, and run reports via natural language voice interface

APPENDICES

Reference Material

Appendix A: Glossary of Terms

AI Agent	A software actor powered by a large language model (LLM) that can autonomously execute tasks within the UME OrgOS, governed by an ODD contract.
DAO	Decentralised Autonomous Organisation. A governance structure in which rules are encoded in smart contracts on a distributed ledger and executed automatically without centralised control.
DLT	Distributed Ledger Technology. A database that is consensually shared and synchronised across multiple sites. UME supports both permissioned (Hyperledger Fabric) and public (Ethereum L2) DLT.
ODD	Organisational Design Document. The versioned, machine-readable contract that defines the capabilities, constraints, and operating rules of any OrgExec within the OrgOS.
OrgApp	An application module running within the UME OrgOS. First-party OrgApps include Finance, HR, Legal, CRM, Operations, Risk, OrgSystem, DAO, Marketing, Work, Portal, and Board.
OrgCPU	An IoT device, physical sensor, industrial machine, or other hardware asset registered as an OrgExec within the OrgOS, operating under its own ODD device contract.
OrgDNA	The exportable, portable configuration of an entire organisation — its structure, module settings, ODD templates, and roles — stored as a versioned artifact in the .umegen format.
OrgExec	Any actor that operates within an OrgOS — a human employee, an AI agent, or an OrgCPU device. All OrgExecs operate under an ODD contract.
OrgGraph	The unified data model of the UME platform. All entities across all modules are nodes in the OrgGraph, connected by typed relationship edges.

OrgKernel	The central operating kernel of the UME platform. Provides event routing, ODD enforcement, state management, audit logging, and module orchestration.
OrgOS	Organisation Operating System. The complete UME platform instance for a single organisation.
RTI	Real Time Information. The HMRC system for payroll reporting in the UK. UME Finance submits FPS and EPS reports via the RTI API.
SDO	Software-Defined Organisation. The real-time digital twin of an organisation maintained by the OrgKernel. Every entity in the organisation is represented as a node in the SDO graph.
Smart Contract	A self-executing program deployed on a distributed ledger that automatically enforces the terms of an agreement when predefined conditions are met.
UQL	UME Query Language. The graph query language used to traverse and query the SDO digital twin.
UPSS	UME Permission Scope Standard. The standard format for capability scopes used in ODD contracts: module:resource:action.

Appendix B: Core API Reference

The following tables document the primary API endpoints for each module. Full OpenAPI 3.1 specifications are available at https://docs.ume.io/api/v3.

B.1 Finance Module API

Method	Endpoint	ODD Scope Required	Description
GET	/v3/fin/invoices	fin:invoice:read	List invoices with filtering, pagination, and sorting
POST	/v3/fin/invoices	fin:invoice:write	Create a new invoice draft
PATCH	/v3/fin/invoices/{id}	fin:invoice:write	Update invoice fields
POST	/v3/fin/invoices/{id}/send	fin:invoice:write	Send invoice to customer (email + Stripe link)
POST	/v3/fin/invoices/{id}/record-payment	fin:invoice:write	Record a payment against an invoice
GET	/v3/fin/journal-entries	fin:je:read	List journal entries
POST	/v3/fin/journal-entries	fin:je:write	Create and post a journal entry
GET	/v3/fin/reports/pl	fin:reports:read	Generate P&L report for a period
GET	/v3/fin/reports/balance-sheet	fin:reports:read	Generate balance sheet as at a date
POST	/v3/fin/payroll-runs	fin:payroll:execute	Create a new payroll run
POST	/v3/fin/payroll-runs/{id}/process	fin:payroll:execute	Process (finalise) a payroll run

B.2 HR Module API

Method	Endpoint	ODD Scope Required	Description
GET	/v3/hr/orgexecs	hr:employee:read	List all OrgExecs (humans, agents, devices)
POST	/v3/hr/orgexecs	hr:employee:write	Create a new OrgExec (new hire)
GET	/v3/hr/orgexecs/{id}/odd	hr:odd:read	Get the current ODD contract for an OrgExec
PUT	/v3/hr/orgexecs/{id}/odd	hr:odd:write	Propose a new version of an ODD contract
POST	/v3/hr/leave-requests	hr:leave:write	Submit a leave request
POST	/v3/hr/leave-requests/{id}/approve	hr:leave:approve	Approve a leave request
POST	/v3/hr/leave-requests/{id}/reject	hr:leave:approve	Reject a leave request with reason

B.3 OrgSystem API

Method	Endpoint	ODD Scope Required	Description
GET	/v3/sys/sdp/graph	sys:sdp:read	Query the SDO graph (UQL or GraphQL body)
GET	/v3/sys/audit	sys:audit:read	Query audit trail with filters
GET	/v3/sys/orgdna	sys:orgdna:read	Get current OrgDNA artifact
POST	/v3/sys/orgdna/export	sys:orgdna:export	Export OrgDNA as .umegen archive
POST	/v3/sys/orgdna/clone	sys:orgdna:write	Clone this OrgDNA to create a new OrgOS instance
GET	/v3/sys/orgapps	sys:orgapp:read	List installed OrgApps and status

Method	Endpoint	ODD Scope Required	Description
POST	/v3/sys/orgapps/{id}/install	sys:orgapp:write	Install an OrgApp from the marketplace

Appendix C: Complete Event Type Reference

The following is the complete list of standard event types emitted by the UME OrgKernel event bus. All events conform to the UME Event Standard schema defined in Section 3.5.

C.1 Finance Events

Event Type	Description	Key Payload Fields
fin.invoice.created	New invoice created (draft or sent)	invoice_id, customer_id, amount, currency, status, due_date
fin.invoice.sent	Invoice dispatched to customer	invoice_id, delivery_method, sent_at
fin.invoice.paid	Full payment received	invoice_id, amount_paid, payment_method, paid_at
fin.invoice.overdue	Invoice passed due date unpaid	invoice_id, days_overdue, amount_outstanding
fin.je.posted	Journal entry posted to GL	je_id, lines[], total_debit, total_credit, period
fin.payroll.processed	Payroll run finalised and processed	run_id, period, employee_count, gross_total, net_total
fin.period.closed	Accounting period locked	period_id, period_name, closed_by, trial_balance_hash
fin.bank.reconciled	Bank account reconciled for period	account_id, period, matched_count, unmatched_count
fin.kri.runway_warning	Cash runway KRI below warning threshold	runway_days, cash_balance, daily_burn_rate, threshold

C.2 HR & ODD Events

Event Type	Description	Key Payload Fields
hr.orgexec.onboarded	New OrgExec activated in the OrgOS	orgexec_id, type, role, odd_version, entity_id

Event Type	Description	Key Payload Fields
hr.orgexec.offboarded	OrgExec deactivated	orgexec_id, offboarding_reason, offboarded_at
hr.odd.published	New ODD version published for an OrgExec	orgexec_id, old_version, new_version, changed_by
hr.leave.requested	Leave request submitted	request_id, orgexec_id, start_date, end_date, leave_type
hr.leave.approved	Leave request approved	request_id, approved_by, approved_at
hr.leave.rejected	Leave request rejected with reason	request_id, rejected_by, reason
hr.payroll.payslip_generated	Payslip generated for employee	payslip_id, orgexec_id, period, net_pay

C.3 CRM & Operations Events

Event Type	Description	Key Payload Fields
crm.order.created	New order created	order_id, customer_id, items[], estimated_value
crm.order.started	Order work commenced	order_id, assigned_orgexec_id, started_at
crm.order.ready	Order ready for collection	order_id, ready_at, customer_notified
crm.order.completed	Order collected or delivered	order_id, completed_at, method
crm.order.invoiced	Invoice generated for completed order	order_id, invoice_id, amount
ops.inventory.low_stock	Inventory item below reorder point	sku_id, current_stock, reorder_point, supplier_id
ops.equipment.alert	OrgCPU device reporting alert condition	device_id, alert_type, reading, threshold, timestamp
ops.po.approved	Purchase order approved	po_id, supplier_id, total_value, approved_by

C.4 DAO & Governance Events

Event Type	Description	Key Payload Fields
dao.proposal.submitted	Governance proposal submitted	proposal_id, submitted_by_org, title, voting_opens_at
dao.vote.cast	Member organisation cast a vote	proposal_id, voter_org_id, vote (for/against/abstain), cast_at
dao.proposal.passed	Governance proposal passed	proposal_id, votes_for, votes_against, execution_contract
dao.royalty.collected	Royalty payment collected from member	contract_id, member_org_id, amount, period, tx_hash
dao.royalty.distributed	Royalty distributed to treasury/parent	contract_id, total_amount, recipients[], tx_hash
dao.smart_contract.deployed	New smart contract deployed to DLT	contract_id, contract_type, deployer_org_id, chain, address
dao.member.joined	New organisation joined the DAO	dao_id, member_org_id, joined_at, orgdna_version

Appendix D: Compliance Framework Reference

D.1 UME Security Certifications & Compliance

Framework / Certification	Status	Scope	Next Review
SOC 2 Type II	In Progress (Q4 2026 target)	All UME Cloud services	Annual
ISO 27001	Planned (2027)	Information Security Management System	Annual
UK Cyber Essentials Plus	Certified	All production systems	Annual
GDPR / UK GDPR	Compliant — DPA available	All EU/UK customer data	Continuous
CCPA	Compliant	US customer personal data	Annual review
PCI DSS	SAQ A (payment page hosted by Stripe/GoCardless)	Payment processing	Annual
HMRC Making Tax Digital	Approved software provider	UK VAT and PAYE	On HMRC software list
FedRAMP	Planned (2027 — US Federal edition)	US Government deployment	Continuous

D.2 Data Residency Options

Region	Residency Guarantee	Data Covered	Certificate
EU (Ireland/Frankfurt)	All data stays within EU boundaries; no US transfer without SCCs	All customer data, backups, logs	EU SCCs available; DPA provided

Region	Residency Guarantee	Data Covered	Certificate
UK (London)	All data stays within UK; GDPR-UK compliant	All customer data, backups, logs	UK IDTA available; DPA provided
US (Virginia/Oregon)	All data stays within US; no international transfer	All customer data, backups, logs	US DPA provided
APAC (Singapore/Sydney)	All data stays within APAC region	All customer data, backups, logs	DPA provided; PDPA compliant

D.3 SLA Reference

Metric	UME Cloud SMB	UME Cloud Enterprise	UME Dedicated
Monthly Uptime	99.95%	99.99%	99.99%
Scheduled Maintenance Window	Sunday 02:00–04:00 UTC	Rolling; zero-downtime deploys	Agreed with customer
Incident Response — Critical (P1)	< 1 hour	< 30 minutes	< 15 minutes + dedicated CSM
Incident Response — High (P2)	< 4 hours	< 2 hours	< 1 hour
Data Backup Frequency	Hourly snapshots + continuous WAL	Continuous + point-in-time recovery	As UME Cloud Enterprise
PITR Window	7 days	30 days	90 days
Support Channels	Email + in-app chat	Email + chat + phone + dedicated CSM	As Enterprise + on-site option
SLA Credit	10% per 0.1% downtime below SLA	25% per 0.01% downtime below SLA	Negotiated

Appendix E: ODD Template Reference

The following tables provide the standard ODD capability grants for common OrgExec role types in a typical SMB organisation.

E.1 Owner / Director ODD

Module	Read	Write	Execute	Restrictions
Finance	✓ All	✓ All	✓ Payroll, Period Close	None
HR	✓ All incl. salary	✓ All	✓ Hire/Offboard	None
Legal	✓ All	✓ All	✓ File, Sign	None
CRM	✓ All	✓ All	✓ All	None
Operations	✓ All	✓ All	✓ All	None
Risk	✓ All	✓ All	✓ Board reports	None
OrgSystem	✓ All	✓ ODD, OrgDNA	✓ Deploy agents	None
DAO	✓ All	✓ All	✓ Vote, Sign contracts	None

E.2 Manager ODD

Module	Read	Write	Execute	Restrictions
Finance	✓ Reports, AR, AP	✓ Invoices, Expenses	✗ Payroll, Period Close	Invoices ≤ £10,000 only
HR	✓ Own dept (no salaries)	✓ Leave approve, Notes	✓ Initiate hire wizard	Cannot see salary data

Module	Read	Write	Execute	Restrictions
Legal	✓ Contracts, Calendar	✓ Contracts < £50k	✗ File with HMRC	Contract value limit enforced
CRM	✓ All	✓ All	✓ All	None
Operations	✓ All	✓ Schedule, POs ≤ £5k	✗ Major POs	PO approval threshold
Risk	✓ Own dept risks	✓ Risk entries	✗ Board reports	Dept-scoped view only
OrgSystem	✓ ODD (own dept)	✗ ODD, OrgDNA	✗ Deploy agents	Read-only system access
DAO	✗	✗	✗	No DAO access

E.3 Standard Employee ODD

Module	Read	Write	Execute	Restrictions
Finance	✗	✗	✗	No finance access
HR	✓ Own record only	✓ Leave request	✗	Self-service only
Legal	✗	✗	✗	No legal access
CRM	✓ Assigned orders	✓ Order status update	✗	Own assigned orders only
Operations	✓ Schedule, Equipment	✓ Consumption record	✗	Own tasks only
Risk	✗	✗	✗	No risk access
OrgSystem	✗	✗	✗	No system access
DAO	✗	✗	✗	No DAO access

Appendix F: OrgDNA JSON Schema Reference

The following defines the complete OrgDNA schema in a structured reference format. The full JSON Schema (Draft 2020-12) is available at https://docs.ume.io/schema/orgdna/v3.

JSON Path	Type	Required	Description
$.schema_version	string (semver)	Yes	OrgDNA schema version — e.g. "3.0.0"
$.orgdna_version	string (semver)	Yes	This OrgDNA artifact version — e.g. "1.4.2"
$.generated_at	string (ISO-8601)	Yes	Timestamp when this version was generated
$.identity.org_id	string (UUID)	Yes	Globally unique organisation identifier
$.identity.display_name	string	Yes	Organisation display name
$.identity.legal_name	string	Yes	Registered legal name
$.identity.jurisdiction	string (ISO 3166)	Yes	Primary jurisdiction — e.g. "GB-ENG"
$.identity.entity_type	enum	Yes	One of: sole_trader, ltd, llc, llp, plc, partnership, nonprofit, dao
$.identity.tax_ids[]	array of objects	No	Tax identification numbers — e.g. UTR, VAT number, EIN
$.structure.departments[]	array of objects	Yes	Department tree with id, name, parent_id, cost_centre
$.structure.reporting_lines[]	array of objects	No	OrgExec reporting relationships
$.modules[].module_code	string (enum)	Yes	Module code — e.g. FIN, HR, LEG, CRM
$.modules[].enabled	boolean	Yes	Whether this module is active

JSON Path	Type	Required	Description
$.modules[].settings	object	No	Module-specific configuration (varies per module)
$.odd_templates[]	array of objects	Yes	ODD contract templates keyed by template_id
$.roles[].role_id	string	Yes	Unique role identifier
$.roles[].odd_template_id	string	Yes	Reference to odd_templates[]
$.compliance.filing_calendar	array of objects	No	Configured filing obligations
$.compliance.vat_config	object	No	VAT scheme, registration number, return frequency
$.compliance.payroll_config	object	No	Payroll frequency, pension scheme, tax regime

8b. Finance Module — Extended Reference

8b.1 Bank Reconciliation Engine

The bank reconciliation engine automates the matching of bank statement transactions against ledger entries. The engine uses a multi-pass matching algorithm:

50. Exact match pass: Match by transaction amount + reference number exact string match. Typically matches 60–75% of transactions automatically.
51. Fuzzy reference match: Match by amount + reference number Levenshtein distance ≤ 2. Captures minor OCR errors or reference truncation. Adds ~10% match rate.
52. Date window match: Match by amount within ±3 business day window. Captures timing differences between banking day and posting date. Adds ~8% match rate.
53. Combination match: Match a single bank transaction to multiple ledger entries that sum to the bank amount (or vice versa). Handles bulk payments.
54. Manual review: Remaining unmatched items are surfaced for manual review with AI-suggested matches ranked by confidence score.

8b.2 Multi-Entity Consolidation

For organisations with multiple legal entities (subsidiaries, group companies, franchise operations), the Finance module provides a full consolidation accounting engine:

Consolidation Step	Process	Output
Chart of Accounts Alignment	Maps each entity's local COA to the group standardised COA using configurable mapping rules	Normalised trial balances per entity in group COA format
Currency Translation	Translates entity financials to the presentation currency using spot rates, average rates, or closing rates per IFRS/UK GAAP rules	FX-translated trial balances with translation reserve
Intercompany Elimination	Identifies and eliminates intercompany balances (loans, trading, dividends) to avoid double-counting in group accounts	Elimination journal entries per entity pair
Minority Interest	Calculates and attributes non-controlling interests for subsidiaries not 100% owned	Minority interest balance sheet and P&L lines

Consolidation Step	Process	Output
Group Aggregation	Aggregates all entity trial balances (after translation and eliminations) into consolidated P&L, Balance Sheet, and Cash Flow	Consolidated financial statements
Variance Analysis	Compares consolidated results against budget, prior period, and prior year	Variance report with commentary prompts for AI-assisted narrative generation

8b.3 Budgeting Module

The budgeting module supports three budgeting methodologies:

- Zero-Based Budgeting (ZBB): Each budget line is built from scratch each period, justified by the operational plan. No automatic roll-forward.
- Incremental Budgeting: Prior period actuals are used as the base, with configurable percentage adjustments per cost category. Fastest for simple SMB use cases.
- Driver-Based Budgeting: Revenue and cost lines are driven by operational metrics (e.g. projected order volume × average order value → revenue; headcount × salary rate → payroll budget). Linked to the Operations and HR modules.

Budget Feature	Description
Rolling Forecast	Monthly reforecast replacing remaining budget with latest estimate; maintains 12-month rolling view
Variance Alerts	Configurable alerts when actuals exceed budget by a threshold (e.g. > 10% adverse on any P&L line)
Department Budgets	Budgets can be set at department level; department heads can view own budget vs actuals only (ODD-scoped)
Budget Approval Workflow	Budget submission → Finance Director review → CEO approval → lock; tracked in ODD approval workflow
Scenario Planning	Three scenario support: Base, Upside, Downside with probability-weighted blended forecast
Budget Import	Import budget from Excel template; auto-maps to COA using fuzzy matching

8b.4 Tax Engine Configuration

The UME tax engine is configured through jurisdiction-specific tax rule packages. The following tax rule packages are available in v3.0:

Tax Package	Jurisdiction	Taxes Covered	Regulatory Integration
UK-STANDARD	England, Wales, Scotland, N. Ireland	VAT (Standard, Reduced, Zero, Exempt), PAYE Income Tax, NIC (Class 1/1A/2/4), Corporation Tax	HMRC MTD API, RTI
UK-CONSTRUCTION	UK	As UK-STANDARD + CIS (Construction Industry Scheme) withholding	HMRC CIS API
EU-VAT	EU Member States	EU VAT including OSS (One Stop Shop) for cross-border digital services	EU OSS portal integration
US-FEDERAL	United States	Federal Income Tax withholding, FICA (Social Security + Medicare), FUTA	EFTPS API, IRS e-file
US-MULTISTATE	US (multi-state)	As US-FEDERAL + state income tax withholding for all 50 states + DC	Per-state agency integration
AU-STANDARD	Australia	GST, PAYG withholding, Superannuation (11.5% SGL)	ATO Single Touch Payroll API
SG-STANDARD	Singapore	GST, CPF contributions (employee + employer)	IRAS API, CPF API

9b. HR Module — Extended Reference

9b.1 Performance Management Framework

The performance management framework supports three review cycle models:

Model	Description	Best For
Annual Review	Single annual review with structured self-assessment, manager assessment, and rating	Traditional SMBs; simpler administrative overhead
Bi-annual Review	Mid-year check-in + end-of-year formal review; ratings only at year-end	Growth-stage businesses; balances rigour with agility
Continuous OKR	Quarterly OKR-setting and review; no single annual rating; always-on feedback	Tech companies; output-focused teams

Each review cycle generates a structured review record with: objectives, key results, self-assessment narrative, manager assessment narrative, 360-degree feedback summary, development priorities, and a formal rating (if applicable). Review records are stored in the HR document vault and linked to the OrgExec record in the SDO.

9b.2 Leave Management Jurisdiction Rules

Jurisdiction	Annual Leave	Sick Pay	Parental Leave	Notes
UK (England/Wales)	28 days statutory (5.6 weeks) incl. bank holidays	SSP £116.75/week from day 4; 28 weeks max	Maternity 52 weeks (39 paid at SMP rate); Paternity 2 weeks; Shared Parental Leave up to 50 weeks	Part-time: pro-rata by contracted hours
Scotland	28 days statutory	As England/Wales	As England/Wales	Different bank holiday dates
United States	No federal minimum; state variations	No federal mandate; state FMLA/PFL	FMLA 12 weeks unpaid (50+ employees)	Configure per-state rules; California, NY differ substantially

Jurisdiction	Annual Leave	Sick Pay	Parental Leave	Notes
European Union	Minimum 4 weeks (EU Working Time Directive)	Varies by member state	EU minimum 4 months parental leave	Member state rules apply; configure per entity jurisdiction
Australia	4 weeks annual leave + 10 days personal/carer's leave	Personal/carer's leave as above (paid)	Parental Leave Pay (PLP) 20 weeks at minimum wage from government	Modern Awards may provide additional entitlements

9b.3 Offboarding Workflow

The offboarding workflow ensures a structured, auditable departure process that protects the organisation legally and operationally:

55. Initiation: HR Manager or Owner initiates offboarding for an OrgExec, specifying: departure reason (resignation/dismissal/redundancy/retirement), last working day, and notice period status.
56. Notice Period Management: If notice is being served, the employee's ODD is updated to reflect any changed access during the notice period (e.g. restricted to read-only on sensitive modules).
57. Knowledge Transfer: Work Management tasks are generated for knowledge transfer activities specific to the departing employee's role.
58. IT Access Revocation: On the last working day, the offboarding workflow triggers immediate ODD suspension across all modules, revocation of all API keys, and SSO session termination.
59. Payroll Final Settlement: Finance module is notified to calculate final pay including accrued but untaken holiday, any PILON (payment in lieu of notice), and statutory payments.
60. Reference Data Retention: Employee record is archived (not deleted) per data retention policy. The OrgExec is moved to Inactive status in the SDO.
61. Legal Checklist: Legal module generates an offboarding legal checklist: return of company property, IP assignment confirmation, confidentiality obligations reminder, reference policy.

13b. Risk Module — Extended Reference

13b.1 Risk Matrix

UME uses a standard 5×5 risk impact × likelihood matrix to score risks. The inherent risk score determines the management response level:

Score	Likelihood	Impact: Low (1)	Impact: Medium (2)	Impact: High (3)	Impact: Critical (4)
5 — Certain	Very likely to occur within 12 months	5 — Medium	10 — High	15 — Critical	20 — Critical
4 — Likely	More likely than not in 12 months	4 — Low	8 — Medium	12 — High	16 — Critical
3 — Possible	~50/50 in 12 months	3 — Low	6 — Medium	9 — High	12 — High
2 — Unlikely	Less than 30% in 12 months	2 — Low	4 — Low	6 — Medium	8 — Medium
1 — Rare	Less than 5% in 12 months	1 — Low	2 — Low	3 — Low	4 — Low

13b.2 GRC Control Library

The GRC module ships with a library of pre-built controls mapped to common frameworks. Controls can be assigned to OrgExecs, linked to evidence documents in the Legal vault, and assessed for operating effectiveness.

Control ID	Control Name	Framework Mapping	Evidence Type	Frequency
CTL-FIN-001	Dual authorisation for payments > £10,000	COSO PCI-06, ISO 27001 A.9.4	Payment approval audit log	Per transaction
CTL-FIN-002	Monthly bank reconciliation completed within 5 working days	COSO AC-12	Reconciliation completion record	Monthly

Control ID	Control Name	Framework Mapping	Evidence Type	Frequency
CTL-FIN-003	Quarterly management accounts reviewed and approved by Director	COSO FM-05	Signed management accounts	Quarterly
CTL-HR-001	All new OrgExecs have ODD contract before system access	ISO 27001 A.9.2.1	ODD publication record	Per hire
CTL-HR-002	ODD contracts reviewed annually	ISO 27001 A.9.2.5	Review completion record	Annual
CTL-IT-001	MFA enabled for all human OrgExecs with financial module access	ISO 27001 A.9.4, CIS v8 #6	Authentication log — MFA usage rate	Continuous monitoring
CTL-IT-002	Penetration test conducted by qualified third party annually	ISO 27001 A.18.2.3, Cyber Essentials+	Pentest report + remediation evidence	Annual
CTL-COM-001	VAT return submitted on time (within 5 days of deadline)	HMRC MTD compliance	Submission confirmation from HMRC API	Quarterly
CTL-COM-002	PAYE RTI Full Payment Submission sent on or before pay day	HMRC RTI compliance	FPS submission confirmation	Monthly

13b.3 Board Risk Reporting

The Risk module generates a standard Board Risk Report suitable for presentation to the board of directors. The report is auto-generated on a configurable schedule (monthly or quarterly) and includes:

- Executive risk summary: current top 5 risks with change indicators (new / increased / decreased / closed) since last report.

- Risk heatmap: 5×5 matrix visualisation showing all open risks by inherent and residual score.
- KRI dashboard: current vs threshold values for all configured KRIs, with trend charts for the past 12 months.
- Compliance calendar: upcoming filing obligations within the next 90 days with status (on track / at risk / overdue).
- Control effectiveness: GRC assessment scores by framework and control category.
- Incident register: any operational incidents in the period with root cause and remediation status.

PART XV

Implementation Guide

24. Implementation Guide

24.1 Implementation Phases

A standard UME implementation for an SMB follows a four-phase approach. Total time from contract signature to full operational deployment is typically 4–8 weeks depending on data migration complexity and organisational readiness.

Phase	Duration	Activities	Deliverables
Phase 1: Foundation	Week 1–2	OrgDNA configuration; entity setup; chart of accounts; role and ODD template setup; SSO integration; data migration planning	Configured OrgOS instance; migrated historical financial data; all OrgExecs with active ODDs
Phase 2: Core Operations	Week 2–4	Finance module go-live (invoicing, GL, payroll); HR module go-live (employee records, leave); CRM order management; Legal compliance calendar	Live financial management; payroll running on UME; order management operational
Phase 3: Extended Modules	Week 4–6	Operations module (inventory, equipment OrgCPUs); Risk module (risk register, KRI config); Work management; Marketing campaigns	Full operational coverage across all modules
Phase 4: AI & Automation	Week 6–8	AI OrgExec deployment (Scheduling, Comms, Compliance Monitor); ODD fine-tuning; SDO digital twin validation; DAO layer (if applicable)	Fully agentic OrgOS; AI agents in production; stakeholder sign-off

24.2 Data Migration Framework

Data migration is typically the most complex element of an OrgOS implementation. UME provides a structured migration framework:

Data Category	Source	Migration Method	Validation Steps
Chart of Accounts	QuickBooks / Xero / Excel	CSV import with COA mapping wizard	Manual review of account mapping; trial balance reconciliation
Historical Transactions	Prior accounting system	CSV import via Finance Migration API (last 24 months recommended)	Trial balance matches source system; sample transaction spot-checks
Customer Records	CRM / spreadsheets	CSV import via CRM Migration API	Record count reconciliation; sample data quality check
Employee Records	HR system / spreadsheets	CSV import via HR Migration API	Headcount reconciliation; ODD provisioning check for all employees
Open Invoices	Prior system	CSV import — AR balance must match GL	AR ageing reconciliation; open invoice count check
Open POs / Bills	Prior system	CSV import — AP balance must match GL	AP ageing reconciliation
Contracts / Documents	Document management / email	Bulk upload to Legal Vault with metadata tagging	Document count; metadata completeness
Bank Balances	Bank statements	Opening balance journal entry + bank statement import from Day 1	Cash balance reconciliation to bank statement

24.3 Cutover Checklist

The following is the standard go-live cutover checklist. All items must be confirmed complete by the implementation lead before activating the new OrgOS for production use:

#	Checklist Item	Owner	Status Field
1	OrgDNA v1.0 published and all modules configured	UME Implementation Lead	Complete / Outstanding
2	All OrgExecs onboarded with active ODD contracts	HR Manager	Complete / Outstanding

#	Checklist Item	Owner	Status Field
3	Chart of accounts mapped and historical data migrated	Finance Director	Complete / Outstanding
4	Opening balance trial balance reconciled to prior system	Finance Director	Complete / Outstanding
5	All customers imported; CRM order management tested end-to-end	Operations Manager	Complete / Outstanding
6	Payroll run tested in staging environment with dummy data	HR Manager	Complete / Outstanding
7	Compliance calendar validated against known filing obligations	Company Secretary / Owner	Complete / Outstanding
8	Bank feeds connected and reconciliation tested	Finance Director	Complete / Outstanding
9	AI OrgExec ODDs reviewed and approved by Owner	Owner	Complete / Outstanding
10	All OrgExecs trained on their module access	UME Implementation Lead	Complete / Outstanding
11	Backup prior system data exported and archived	IT / Owner	Complete / Outstanding
12	Emergency rollback plan documented and tested	UME Implementation Lead	Complete / Outstanding

PART XVI

Infrastructure Deep Dive

25. Infrastructure Architecture — Deep Dive

25.1 Kubernetes Cluster Architecture

The UME Cloud production deployment runs on a multi-zone Kubernetes cluster per region. Each cluster is structured into dedicated node pools by workload type:

Node Pool	Instance Type	Min / Max Nodes	Workloads	Taints / Labels
orgkernel-pool	c5.2xlarge (8vCPU, 16GB)	3 / 12	OrgKernel services, ODD Engine, Event Bus consumers	critical=true
modules-pool	c5.xlarge (4vCPU, 8GB)	4 / 40	All OrgApp module services	workload=modules
ai-pool	g4dn.xlarge (4vCPU, 16GB, T4 GPU)	0 / 8 (scale-to-zero)	AI Agent runtime containers	gpu=true; workload=ai
data-pool	r5.2xlarge (8vCPU, 64GB)	3 / 6 (stateful)	Redis, Elasticsearch, TimescaleDB sidecars	data=true; NoSchedule others
batch-pool	c5.large (2vCPU, 4GB)	0 / 20 (scale-to-zero)	Batch jobs, report generation, bulk imports	workload=batch
ingress-pool	c5.large (2vCPU, 4GB)	3 / 6	Kong API Gateway, Nginx ingress controllers	ingress=true

25.2 Network Security Architecture

The UME platform network is structured in multiple security zones with strict ingress/egress control between zones:

Zone	Contents	Ingress Allowed From	Egress Allowed To
DMZ (Public)	CDN PoPs, Load Balancers, WAF	Internet (port 443 only)	API Gateway Zone
API Gateway Zone	Kong API Gateway, Rate Limiter, Auth middleware	DMZ	OrgKernel Zone
OrgKernel Zone	OrgKernel services, ODD Engine, Event Bus	API Gateway Zone, Module Zone (gRPC)	Module Zone, Data Zone
Module Zone	All OrgApp module services	OrgKernel Zone	OrgKernel Zone, Data Zone, External Integrations Zone
Data Zone	PostgreSQL, Redis, Neo4j, Kafka, Elasticsearch	OrgKernel Zone, Module Zone	S3 Zone (backups)
S3 Zone	Object storage — documents, backups, audit archive	Data Zone, Module Zone (documents)	No egress to internet; cross-region replication only
External Integrations Zone	Outbound webhooks, HMRC API, Stripe API, Twilio API	Module Zone	Internet (specific allow-listed domains only)
Management Zone	CI/CD pipelines, monitoring, alerting	VPN only	All zones (read/deploy only)

25.3 CI/CD Pipeline Architecture

UME uses a trunk-based development model with automated testing gates at every stage of the deployment pipeline:

62. Code commit to main branch → automated unit test suite (Go/Node.js test frameworks; minimum 85% coverage gate).
63. PR merge → integration test suite runs against a full ephemeral UME test environment (spun up on demand by the pipeline).
64. SAST scan (CodeQL) + dependency vulnerability scan (Dependabot/Snyk) → critical/high vulnerabilities block promotion.

65. Container image build + sign (cosign) → image pushed to private container registry with immutable tag.
66. Staging deployment → automated smoke tests + E2E Playwright tests against staging environment.
67. Performance benchmark: API response time regression test — if p95 degrades by > 15% vs baseline, deployment is blocked.
68. Production deployment using Argo CD GitOps → progressive rollout (canary: 5% → 25% → 100% over 30 minutes with automatic rollback on error rate spike).
69. Post-deployment synthetic monitoring validation — if Playwright canary fails within 10 minutes, automatic rollback triggered.

Appendix G: Worked Example — Bob's Dry Cleaning Ltd

This appendix provides a complete worked example of a UME OrgOS implementation for a small dry-cleaning business, illustrating how the platform concepts described in this document apply in a real-world context.

G.1 Business Overview

Business Name	Bob's Dry Cleaning Ltd
Legal Form	Private Limited Company (Ltd) — England & Wales
Company Number	12345678 (Companies House)
VAT Number	GB 987 654 321
Employees	7 full-time, 2 part-time
Annual Revenue	Approximately £340,000
OrgOS Edition	Growth Plan
Modules Active	Finance, HR, Legal, CRM, Operations, Risk, OrgSystem, Work
AI Agents	Scheduling Agent (AI-SCHED-v2.0), Customer Comms Agent (AI-COMMS-v1.5), Compliance Monitor (AI-COMP-v1.1)

G.2 OrgExec Registry

Name	Role	Type	ODD Template	Key Capabilities
Bob A.	Owner / Director	Human	OWNER-ODD-v1.2	Full access — all modules; DAO voting rights; payroll approve; period close
Alice L.	Manager	Human	MANAGER-ODD-v1.1	All modules; invoice ≤ £10k; PO ≤ £5k; leave approve; no salary view
Tom W.	Senior Cleaner	Human	FTE-CLEANER-ODD-v1.2	CRM orders (own); Operations schedule; HR own record
Sarah P.	Cleaner	Human	FTE-CLEANER-ODD-v1.1	CRM orders (own); Operations schedule; HR own record

Name	Role	Type	ODD Template	Key Capabilities
Mike J.	Driver	Human	FTE-DRIVER-ODD-v1.0	CRM orders (collection/delivery); HR own record; no ops equipment
Scheduling Agent	AI Scheduler	AI Agent	AI-SCHED-ODD-v2.0	CRM read; HR schedule write; Ops schedule write; max 200/hr; escalate at +40% demand
Comms Agent	AI Comms	AI Agent	AI-COMMS-ODD-v1.5	CRM order read, customer notify; SMS/email outbound only; no write; escalate on complaint
Compliance Agent	AI Compliance	AI Agent	AI-COMP-ODD-v1.1	Risk KRI read; Legal calendar read; all read-only; escalate Critical KRI to Bob
Washer A	Industrial Washer	OrgCPU	ORGCPU-WASHER-v1.0	ops:equipment:telemetry; ops:cycle:report; no commands
Washer B	Industrial Washer	OrgCPU	ORGCPU-WASHER-v1.0	ops:equipment:telemetry; ops:cycle:report; no commands

G.3 Typical Daily OrgOS Activity

The following illustrates a typical day of OrgOS activity and how the platform components interact:

Time	Actor	Action	Platform Response
06:00	Scheduling Agent	Reviews overnight order queue (12 orders); optimises schedule for the day	ODD checked (200/hr limit OK); schedule written to Ops; HR notified of assignments; SDO updated
07:30	Tom W.	Logs in; views assigned orders on Order Board	JWT issued; ODD resolved (FTE-CLEANER); orders filtered to Tom's assignments only
09:15	Washer A (OrgCPU)	Completes cycle for order #4521; emits ops.equipment.cycle_complete	OrgKernel routes event; CRM updates order to "Ready"; Comms Agent triggered
09:16	Comms Agent	Sends "Your order is ready" SMS to Mrs Chen	ODD checked (notify:sms permitted); SMS sent via Twilio; audit record created; CRM customer record updated with communication log

Time	Actor	Action	Platform Response
10:30	Alice L.	Raises invoice INV-1086 for £340 and sends to customer	ODD checked (fin:invoice:write; £340 < £10k limit OK); invoice created; SendGrid email dispatched; AR updated; SDO updated
11:45	Compliance Agent	KRI check — cash runway now 18 days, below 30-day threshold	KRI breach event emitted; risk.kri.runway_breach event; Bob notified (push + email); Risk register auto-updated
12:00	Bob A.	Receives KRI breach notification; reviews cash position in Finance	Full Finance module access (Owner ODD); cash flow forecast viewed; AR ageing reviewed
15:00	Bob A.	Approves July payroll run (£21,420 net)	ODD checked (fin:payroll:execute OK); payroll processed; journal entries posted; payslips generated; HR notified; RTI FPS queued for HMRC
17:00	Sarah P.	Submits leave request for Aug 5–7 (3 days)	ODD checked (hr:leave:write OK); leave request created; Alice notified for approval; team coverage check passes (>60%)
17:05	Alice L.	Approves Sarah's leave request	ODD checked (hr:leave:approve OK); leave approved; HR record updated; Finance accruals updated; Sarah notified

Appendix H: Sample ODD Contract — JSON Format

The following is an abridged example of an ODD contract JSON document for the Customer Comms AI Agent. Full JSON Schema is available at https://docs.ume.io/schema/odd/v3.

> Note: The below is a human-readable representation. The actual ODD stored in UME includes all fields specified in Appendix F and is digitally signed by the approving OrgExec.

JSON Field	Value	Explanation
odd_id	"ODD-0007"	Unique ODD identifier in the OrgOS
odd_version	"1.5.2"	Semantic version of this ODD contract
identity.actor_id	"AGENT-COMMS-001"	System identifier for this AI agent
identity.type	"ai_agent"	OrgExec type: human | ai_agent | orgcpu
identity.display_name	"Customer Comms Agent"	Human-readable name shown in OrgSystem
identity.model	"claude-3-5-sonnet"	LLM model this agent runs on
capabilities[0].scope	"crm:order:read"	Can read CRM order data
capabilities[1].scope	"crm:customer:read"	Can read CRM customer records
capabilities[2].scope	"notify:sms"	Can send outbound SMS via Twilio
capabilities[3].scope	"notify:email"	Can send outbound emails via SendGrid
constraints.rate_limit.max_per_hour	500	Maximum 500 notification actions per hour
constraints.rate_limit.max_per_day	5000	Hard daily limit on all outbound messages
constraints.data_scope	"crm.orders:assigned_to_org"	Can only see orders belonging to its own org

JSON Field	Value	Explanation
constraints.operating_hours	"00:00-23:59"	Operates 24/7 (customers may query at any time)
escalation.trigger_conditions[0]	"customer.sentiment == negative"	Escalate to assigned OrgExec if negative sentiment detected
escalation.trigger_conditions[1]	"message_type == complaint"	Always escalate complaints — no AI response
escalation.target_actor_id	"EMP-0002"	Escalates to Alice L. (Manager)
audit.verbosity	"maximum"	Log all LLM reasoning traces and tool calls
audit.retention_years	2	Reasoning logs retained 2 years; notifications 7 years
approved_by	"EMP-0001"	Bob A. (Owner) approved this ODD version
approved_at	"2025-09-01T14:32:00Z"	ISO-8601 approval timestamp
digital_signature	"SHA256:3a7f..."	ECDSA signature of Owner's ODD approval key

Appendix I: Platform Configuration Reference

I.1 OrgKernel Environment Variables

The following environment variables configure the OrgKernel in all deployment models. These are set in the Kubernetes ConfigMap and Secret resources managed by the UME Helm chart.

Variable	Default	Description
ORGKERNEL_ODD_CACHE_TTL_SEC	300	ODD resolved permission cache TTL in seconds
ORGKERNEL_EVENT_BUS_BROKERS	(required)	Comma-separated Kafka broker addresses
ORGKERNEL_EVENT_BUS_TOPIC_PREFIX	ume.events	Kafka topic prefix for all event types
ORGKERNEL_AUDIT_DB_URL	(required)	PostgreSQL connection string for audit log database
ORGKERNEL_AUDIT_S3_BUCKET	(required)	S3 bucket name for long-term audit archive
ORGKERNEL_SDO_NEO4J_URI	(required)	Neo4j Bolt URI for SDO graph database
ORGKERNEL_REDIS_URL	(required)	Redis connection URL for caching and locks
ORGKERNEL_JWT_SECRET_KID	(required)	Key ID for JWT signing key in KMS
ORGKERNEL_MAX_REQUEST_SIZE_MB	50	Maximum inbound request body size in MB
ORGKERNEL_RATE_LIMIT_WINDOW_SEC	60	Rate limit sliding window size in seconds
ORGKERNEL_TRACE_SAMPLE_RATE	0.05	OpenTelemetry trace sample rate (5% default)
ORGKERNEL_LOG_LEVEL	info	Log verbosity: debug | info | warn | error
ORGKERNEL_MULTI_REGION_ENABLED	false	Enable cross-region event replication

Variable	Default	Description
ORGKERNEL_ORGDNA_S3_BUCKET	(required)	S3 bucket for OrgDNA artifact storage
ORGKERNEL_AI_AGENT_REASONING_LOG	true	Whether to capture LLM reasoning traces in audit log

I.2 Module Feature Flags

Feature flags control the availability of specific module capabilities. Flags are stored in the OrgDNA and can be toggled per organisation through the OrgSystem module.

Flag Key	Module	Default	Description
fin.multi_currency.enabled	Finance	false	Enable multi-currency invoicing and GL entries
fin.auto_reconciliation.enabled	Finance	true	Enable automated bank reconciliation matching
fin.consolidation.enabled	Finance	false	Enable multi-entity P&L consolidation (requires Growth+)
hr.360_feedback.enabled	HR	false	Enable 360-degree peer feedback in performance reviews
hr.ai_interview_assist.enabled	HR	false	Enable AI-assisted interview question generation (beta)
crm.micro_persona.enabled	CRM	true	Enable customer micro-persona engine
crm.comms_ai.enabled	CRM	false	Enable AI Comms Agent (requires AI agent ODD)
ops.orgcpu.enabled	Operations	false	Enable IoT OrgCPU device registration and telemetry
risk.board_report_ai.enabled	Risk	false	Enable AI-generated board report narrative (beta)
leg.contract_ai.enabled	Legal	false	Enable AI-assisted contract drafting (beta)

Flag Key	Module	Default	Description
sys.sdp_public_graph.enabled	OrgSystem	false	Enable SDO graph API access for all OrgExecs (default: admin only)
dao.dlt_type	DAO	"fabric"	DLT type: fabric | ethereum_l2 | none
mkt.ai_content.enabled	Marketing	false	Enable AI content generation for campaigns

I.3 Webhook Event Subscription Reference

The following event types are available for outbound webhook subscription. Configure subscriptions through the OrgSystem module or via the /v3/sys/webhooks API.

Category	Available Event Types
Finance	fin.invoice.created, fin.invoice.sent, fin.invoice.paid, fin.invoice.overdue, fin.invoice.cancelled, fin.je.posted, fin.payroll.processed, fin.period.closed, fin.period.opened, fin.bank.reconciled, fin.budget.variance_breach
HR	hr.orgexec.onboarded, hr.orgexec.offboarded, hr.orgexec.role_changed, hr.odd.published, hr.odd.approved, hr.leave.requested, hr.leave.approved, hr.leave.rejected, hr.payroll.payslip_generated, hr.performance.review_completed
Legal	leg.filing.due_soon, leg.filing.overdue, leg.filing.completed, leg.contract.created, leg.contract.signed, leg.contract.expired, leg.compliance_score.changed, leg.entity.created
CRM	crm.order.created, crm.order.started, crm.order.ready, crm.order.completed, crm.order.invoiced, crm.order.closed, crm.order.cancelled, crm.customer.created, crm.customer.updated
Operations	ops.inventory.low_stock, ops.inventory.reorder_triggered, ops.po.created, ops.po.approved, ops.po.received, ops.equipment.alert, ops.equipment.offline, ops.equipment.maintenance_due
Risk	risk.item.created, risk.item.updated, risk.item.closed, risk.kri.warning, risk.kri.breach, risk.incident.created, risk.incident.closed, risk.grc.assessment_completed
OrgSystem	sys.orgapp.installed, sys.orgapp.updated, sys.orgdna.updated, sys.odd.published, sys.odd.approved, sys.ai_agent.deployed, sys.ai_agent.escalation, sys.audit.security_event

Category	Available Event Types
DAO	dao.proposal.submitted, dao.proposal.passed, dao.proposal.failed, dao.vote.cast, dao.royalty.collected, dao.royalty.distributed, dao.member.joined, dao.member.suspended

UME Platform

Organisation Operating System

Design & Architecture Document · v3.0.0 · March 2026

https://ume.io · docs.ume.io · support@ume.io

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